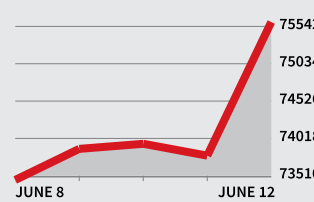


the hindu businessline

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IN FOCUS

	Week's close	Week's change
Nifty 50	23622.90	+256.20
P/E Ratio (Sensex)	20.41	+0.32
US Dollar (in ₹)	95.11	+0.17
Gold Std 10 gm (in ₹)	147208.00	-6412
Silver 1 kg (in ₹)	242582.00	-14326

LOGISTICS.

As the country's first ship recycling credit note goes out, domestic yards hope it will lead to more business p8



BRANDLINE.

Brands of Bengal: 90 Not Out chronicles Bengal's enduring entrepreneurial legacy **p6**

BENGALURU - CHENNAI - COIMBATORE - HUBBALLI - HYDERABAD - KOCHI - KOLKATA - MADURAI - MALAPPURAM - MANGALURU - MUMBAI - NOIDA - THIRUVANANTHAPURAM - TIRUCHIRAPPALLI - VIJAYAWADA - VISAKHAPATNAM

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MONDAY SPECIALS.

PULSE

A widening crisis for Indian women



Women in Indian cities appear to be carrying more than their share of weight. The latest National Family Health Survey shows that nearly 43 per cent of urban women are overweight or obese — a steep climb from 33 per cent in the last survey. At this pace, India is on the brink of a worrying demographic moment when half of all the urban women aged 15–49 will have a BMI above 25. **p9**

QUANTUM

The right time to deploy thorium is now

India's vast thorium resources offer a means for the country to become a global clean energy provider rather than remaining a major energy importer with the attendant vulnerabilities. As demand outstrips uranium supply in the near future, India's vast thorium resources will come into their own. **p7**

Wheels turn for \$10 billion US tariff refunds to India

MONEY TRAIL. RBI eases banking hurdles; digital payment providers offer alternative channels

Amiti Sen
New Delhi

The process of repatriating more than \$10 billion in US reciprocal tariff refunds to Indian exporters has begun, following detailed administrative guidelines issued by US Customs and Border Protection (CBP) and a key clarification from the Reserve Bank of India (RBI) aimed at facilitating the inflows.

Cross-border digital payment providers have also entered the fray, offering exporters mechanisms to receive refunds without opening foreign bank accounts and at significantly lower costs than traditional commercial banking channels, a source familiar with the developments told *businessline*.

PAYOUT PROCESS

However, exporters that paid the duties through freight forwarders face an additional complication. Since the forwarders acted as the legal importer on record, they will receive the refunds directly from the US Treasury and will subsequently have to transfer the funds to



TARIFF FALLOUT. The refunds arise from tariffs levied under a US enforcement initiative launched in April 2025, targeting countries with large trade surpluses with that country

the exporters in compliance with applicable regulations. The operational details of this process are still being worked out, the source said.

In cases where duties were paid directly by US importers, the refunds will be credited to the importing entities. Given that many Indian exporters had initially offered substantial discounts to share the tariff burden, industry expectations are that the refunds will also be shared. “The mechanics of how the two parties divide the refunds will vary on a case-by-case basis,” the source noted.

“The systems are now fall-

ing into place. Delays occurred because the US Customs took time to issue the necessary administrative guidelines. With those now in place, some smaller remittances have already been processed. However, larger payouts have not yet begun and are expected to be released gradually,” the source added.

COURT REPRIEVE

The refunds arise from a broad trade enforcement action launched by Washington in April 2025 under the International Emergency Economic Powers Act (IEEPA), targeting countries

SEBI board may restore open market buybacks

Akshata Gorde
Mumbai

The Securities and Exchange Board of India (SEBI) is expected to clear a fresh set of ease-of-doing-business measures at its June 19 board meeting, including the reintroduction of open-market buybacks through stock exchanges, a faster rollout mechanism for alternative investment fund (AIF) schemes, and simplified transmission norms for legal heirs.

Other proposals likely to come up before the board include permitting online bond platform providers to distribute products regulated by the International Financial Services Centres Authority (IFSCA) and 54EC capital gains tax-saving bonds. The regulator may also consider changes to the pre-open call auction mechanism for IPOs and re-listed companies to improve opening price discovery, according to people familiar with the matter.

Among the key agenda items is the proposed revival of the stock exchange route for open-market buybacks, which was phased out in 2023 amid tax-related concerns.

Details p2

India throws open PIS route to all non-resident investors

Shishir Sinha
New Delhi

India has widened access to its equity markets by allowing all non-resident individuals — not just Non-Resident Indians (NRIs) and Overseas Citizens of India (OCIs) — to invest in listed companies under the Portfolio Investment Scheme (PIS), as part of efforts to deepen capital markets and attract a broader pool of global investors.

A June 12 amendment by the Finance Ministry to the Foreign Exchange Management (Non-Debt Instruments) Rules removes a long-standing restriction that limited PIS investments to NRIs and OCIs.

The change opens the route to foreign retail and institutional investors who previously had limited direct access to Indian listed equities.

“The amendment will broaden the investor base for listed companies over time. The higher investment caps will also benefit NRIs, who were earlier restricted from investing more than 5 per cent in a listed entity under the PIS route,” said Natasha Treasurywala, Senior Partner, Desai & Diwanji.

However, there is a rider. Investments that result in

the transfer of ownership or control of a listed Indian company to entities or citizens of countries sharing a land border with India — or where the beneficial owner is a citizen of such a country — will continue to require prior government approval.

Abir Lal Dey, Partner at CMS INDUSLAW, said the amendment retains regulatory scrutiny of beneficial ownership and control, reflecting the intent of Press Note 3. “The net effect is a calibrated opening: Capital is welcome, but only on terms that preserve regulatory oversight and safeguard national security,” he said.

MIXED VIEWS

Sonam Chandwani, Managing Partner at KS Legal & Associates, said the revised framework adopts a more liberal approach to cross-border investment while retaining investment caps, FDI conversion requirements upon breach of limits, and mandatory government approval in sensitive cases.

Some experts, however, question whether the move will materially boost foreign inflows. Mayank Arora, Associate Partner (Regulatory) at Nangia Global, said the decision appears aimed at reviving foreign investment at a time when both FDI and FPI flows have softened.

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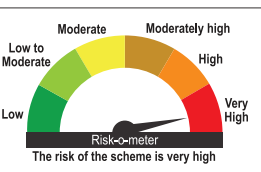
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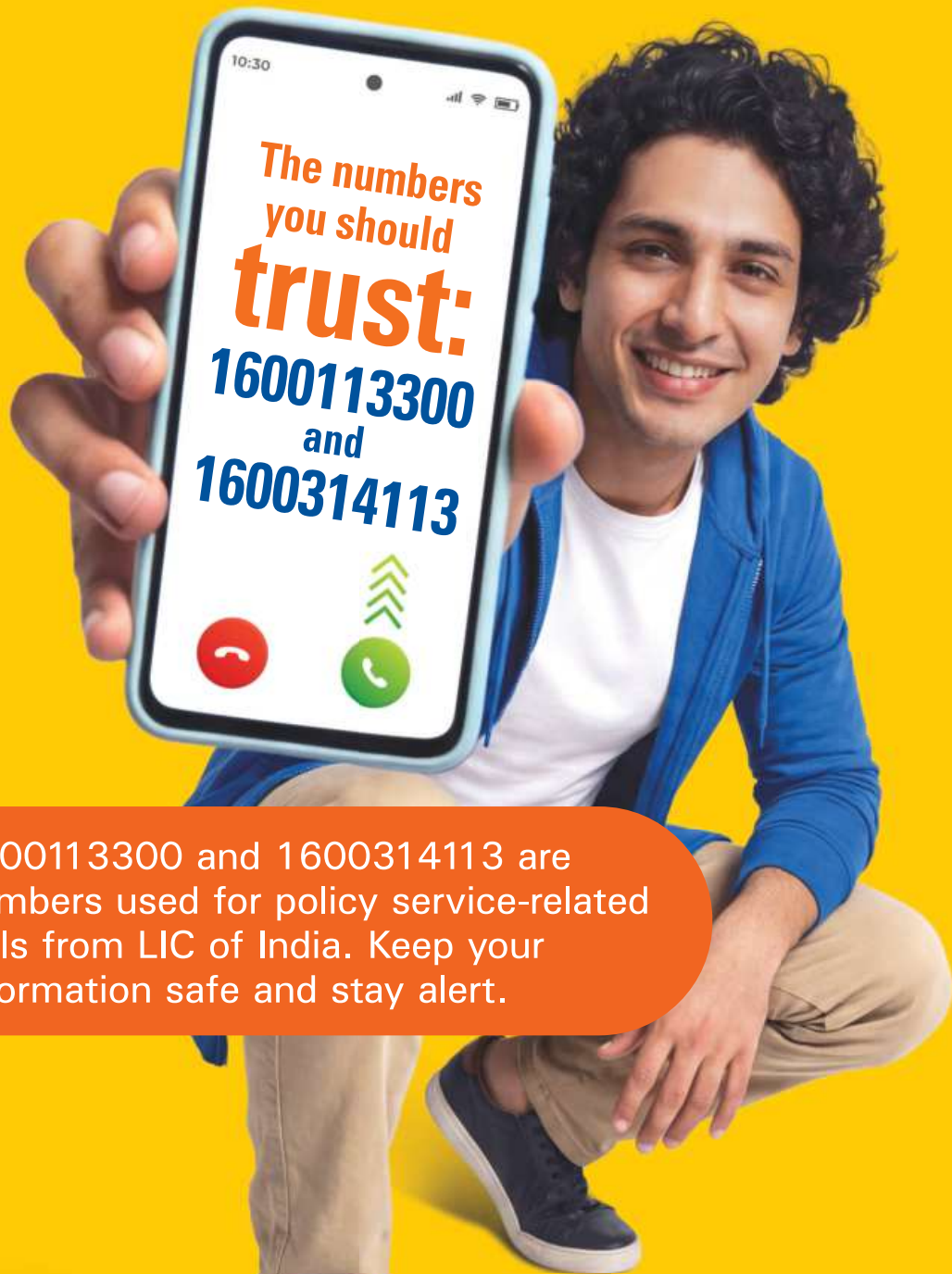


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QUICKLY.

LIC targets stronger market leadership

New Delhi: LIC CEO and MD R Doraiswamy said despite rising competition, the company will stay focused on strengthening its leadership position while contributing to national development. LIC commands close to 60 per cent market share in the life insurance segment and manages assets of over ₹57 lakh crore. It has real estate valued around ₹60,000 crore. The story of LIC, which completed the 70th year of its existence, is inseparable from the story of India. “As the economy grows, LIC grows. LIC growth helps the economy grow. So, it has been a synergistic kind of relationship that LIC has been having,” he said.

No fresh Nipah cases, says Kerala Health Minister

Kozhikode: Kerala Health Minister K Muralreedharan said no fresh Nipah cases have been reported and defended the government's handling of the outbreak. He also said public panic was unwarranted. Speaking to reporters after a review meeting on the Nipah situation, the Minister said the lone Nipah patient had been administered Ribavirin, monoclonal antibody treatment and Remdesivir.

SEBI board may revive open market buybacks on June 19

TOP PRIORITY. Proposal expected to be among the most significant items on the agenda

Akshata Gorde
Mumbai

The Securities and Exchange Board of India (SEBI) is expected to clear a fresh set of ease-of-doing-business measures at its June 19 board meeting, including the reintroduction of open-market buybacks through stock exchanges, a faster rollout mechanism for alternative investment fund (AIF) schemes, and simplified transmission norms for legal heirs.

Other proposals likely to be brought before the board include permitting online bond platform providers to distribute products regulated by the International Financial Services Centres Authority (IFSCA) and 54EC capital gains tax-saving bonds.

The regulator may also consider changes to the pre-open call auction mechanism for IPOs and re-listed companies to improve opening price discovery, according to people familiar with the matter. Among the key agenda

ON THE AGENDA

- Reduce the timeline for launching eligible AIF schemes to around 10 working days
- Simplify transmission of securities by doubling thresholds to ₹10 lakh for physical holdings and ₹30 lakh for demat holdings
- Revise the base price and price-band framework for ETFs to improve price discovery

items is the proposed revival of the stock exchange route for open-market buybacks, which was phased out in 2023 amid tax-related concerns.

BUYBACK SAFEGUARDS Through consultation papers issued in April and May, SEBI proposed reintroducing the stock exchange route with tighter safeguards.

These include shortening the buyback window to 66 working days from six months, mandating utilisation of at least 40 per cent of

the buyback size in the first half of the offer period, enhancing disclosures and restricting purchases from promoters and promoter-group entities during the buyback.

The regulator also proposed streamlining buyback regulations by removing redundant provisions, aligning the framework with the T+1 settlement cycle, simplifying procedures and reducing compliance timelines for listed companies.

FASTER AIF LAUNCHES Another major proposal is

Garuda (Green-Channel: AIF Rollout Upon Document Acknowledgement), which is aimed at cutting the time that is required to launch AIF schemes from the current fast-track window of 30 days.

The mechanism could reduce the approval timeline to around 10 working days and, in certain cases such as accredited-investor-only or angel funds, allow near-immediate launches upon acknowledgment.

The board is also expected to consider doubling the monetary thresholds for simplified transmission of securities to ₹10 lakh for physical holdings and ₹30 lakh for demat holdings, while easing documentation requirements for legal heirs and nominees.

Separately, SEBI has proposed revising the base-price and price-band framework for ETFs by allowing exchanges to use a dynamic reference price linked to the indicative net asset value (iNAV).

An email sent to SEBI did not receive a response till press time.

‘Differential fuel pricing may play a role in shaping adoption patterns’

S Ronendra Singh
New Delhi

The country's second largest two-wheeler manufacturer, Honda Motorcycle and Scooter India (HMSI), has said that differential fuel pricing is among the primary factors that will play a role in the development of the flex fuel ecosystem.

“Pricing dynamics, fuel availability and overall market conditions are among the factors influencing the development of the flex-fuel ecosystem. Continuous technology development and optimisation are expected to further enhance overall customer value,” Tsutsumu Otani, President and Chief Executive Officer, HMSI, told *businessline*.

On the recent exemption of excise duty on flex-fuel vehicles, Otani said any potential impact of excise duty changes would depend on how these factors evolve together and how end customers ultimately benefit within the overall ecosystem.

On customers complaining about the impact on mileage because of ethanol blended fuel, he said fuel efficiency outcomes may vary depending on blend levels,



Tsutsumu Otani, President and CEO, HMSI

vehicle technology and actual driving conditions.

“As a result, cost per kilometre can differ based on fuel prices, usage patterns and operating environments. Compatibility with higher ethanol blends depends on the specific vehicle design and its intended operating parameters,” Otani explained, adding that vehicles engineered for flex-fuel applications are designed and validated accordingly, while others may require appropriate technical adaptation, depending on their architecture.

FLEX-FUEL PUSH Though the top official at

HMSI did not share any timeline of launching a flex-fuel vehicles, according to sources, the company is expected to launch a flex-fuel two-wheeler next month.

HMSI previously (in 2024) had launched CB300F flex-fuel motorcycle, which could run on E85 fuel, but the model was discontinued later because of weak demand.

“The transition toward lower-emission mobility is expected to be supported by multiple technology pathways, including advancements in internal combustion engine (ICE) efficiency and the use of alternative fuels. Flex-fuel vehicles represent one of the potential approaches within this broader mobility landscape, alongside other evolving solutions,” Otani added.

EXCISE RELIEF Last week, the Finance Ministry issued a series of notifications exempting petrol with an ethanol blend of 22 per cent to 30 per cent from central excise duty.

However, the Oil Ministry noted that these higher blends will not be rolled out until extensive testing and stakeholder consultations are complete.

Cockroach Janata Party holds protests in Hyderabad, Bengaluru

Our Bureau
Hyderabad/Bengaluru

As part of its campaign seeking accountability for the NEET paper leak and irregularities related to the CBSE's On-Screen Marking (OSM) system, the newly-formed Cockroach Janata Party (CJP) held protests in Hyderabad and Bengaluru on Sunday. Protesters demanded the resignation of Union Education Minister Dharmendra Pradhan.

The protest at Dharna Chowk in Hyderabad was attended by more than 1,000

people, including education reformer and activist Sonam Wangchuk, political analyst Professor K Nageshwar and CJP's official spokesperson Vijetha Dahiya.

In response to protesters demanding that Wangchuk be made the education minister, he said his struggle is only against the “injustice” done to students and that he is not interested in securing any position of power.

Hyderabad-based activist Ruchit Kamal presided over the gathering. “The reason we can stand here today and protest is because of the sacrifices made by the people

who fought for an independent India. We take inspiration from the martyrs of the Telangana movement.”

BENGALURU The protest at Bengaluru was held amid heavy rain at Freedom Park and was attended by a few hundred demonstrators. Celebrities, including actor Prakash Raj, Kiran Srinivas and wildlife filmmakers Krupakar BS and Senani Hegde, along with environmentalist Nagesh Hegde, extended support.

Some Left and Dalit organisations also extended support to the protest.

May tractor sales top 1 lakh units for 3rd straight month

TE Raja Simhan
Chennai

The tractor industry continued its strong growth trajectory in 2026, with domestic sales crossing the one lakh-unit mark for the third consecutive month in May, supported by favourable rural sentiment and improved farm economics even as monsoon progress remains a key variable to watch for.

According to data from the Tractor and Mechanization Association (TMA), domestic tractor sales stood at 1,08,229 units in May 2026 — up 20 per cent from 90,500 units in the corresponding month last year.

The industry has seen sustained growth through the

year, with monthly sales rising from 88,522 units in January to 1,03,193 units in March, and further to 1,05,021 units in April before touching 1,08,229 units in May.

EXPORTS RECOVER Exports also showed signs of recovery, crossing the 10,000-unit mark for the first time this year.

Tractor exports rose to 10,165 units in May, the highest since March 2025 when overseas shipments stood at 10,733 units.

Prior to that, exports had exceeded the 10,000-unit threshold in March 2024, at 10,794 units, according to TMA data.

Upward trajectory			
Month	Year		Growth Y-o-Y (%)
	2026	2025	
January	88,522	61,923	43
February	78,901	58,797	34
March	1,03,193	79,946	29
April	1,05,021	82,839	27
May	1,08,229	90,500	20

Source: Tractor and Mechanization Association

“A 20 per cent year-on-year growth, supported by the GST reduction on tractors, which continues to improve rural affordability, reflects a healthy demand environment. The fact that absolute volumes have remained above the one lakh-mark for three consecutive months reinforces that trend,” said Poonam Upad-

hyay, Director, Crisil Ratings.

She noted that while the pace of month-on-month growth had moderated, sales continued to expand despite a higher base.

“The GST rate cut came into effect in late September 2025. Consequently, year-ago volumes from that period onwards already in-

corporate the affordability benefit, making second-half comparisons progressively less favourable. Monsoon progression and its impact on farm incomes remain the key variables to watch through the rest of FY27,” she added.

Kubota Ltd, in its monthly bulletin, said the growth momentum continued in May, with healthy performance across wholesale and retail segments. Near-term headwinds include rising input costs, particularly fertilizers, softer prices for select cash crops, and evolving geopolitical developments, which could affect customer affordability and input availability ahead of the kharif season, the company said.

IpcA, Bhami Research team up for monoclonal antibodies

Our Bureau
Mumbai

IpcA Laboratories has inked an agreement with the Mangaluru-headquartered Bhami Research Laboratories for access to a biologics delivery platform to develop monoclonal antibodies.

The licencing agreement grants IpcA access to BRL's proprietary high-concentration subcutaneous biologics delivery platform for the development and commercialisation of multiple monoclonal antibody products targeting oncology and inflammatory diseases, a note from the company said.

LICENSING FEE Ratnesh Jain, Co-founder

with WeTranslate, a life sciences business development company that played a role in the scientific and financial match-making between the two companies, told *businessline* that IpcA would undertake the development, and there was a licensing fee for the technology transfer and royalties on sales, once the product is commercialised.

The financial details of the agreement were not disclosed.

Pranay Godha, MD and CEO, IpcA Laboratories, said that the collaboration would help accelerate the development of “affordable biologics for people battling cancer and autoimmune disease worldwide”.

Dana-Eaton \$5.1 billion deal set to reshape India’s EV, CV supply chain

Amit Vijay Mohile
Mumbai

Days after Dana Inc and Eaton Corporation announced a \$5.1 billion deal to combine their mobility businesses, industry experts said the transaction could have significant implications for India's commercial vehicle (CV) and electric vehicle (EV) supply chain, potentially accelerating localisation, integrated powertrain development and electric mobility adoption.

The deal combines Dana's strengths in axles, drivshafts, thermal-management systems and e-axes with Eaton Mobility's expertise in transmissions, clutches and electrification technologies, creating a sup-

plier with more than \$11 billion in annual revenue and a sizeable presence in India's automotive manufacturing ecosystem.

Industry experts told *businessline* that the combination reflects a broader shift underway in the global automotive sector, from supplying individual components to delivering complete vehicle systems.

SYSTEM INTEGRATION For Indian vehicle manufacturers, the significance lies less in the size of the transaction and more in what it brings together. “This is not merely a merger of two component suppliers. It is effectively the creation of a systems supplier at a time when OEMs increasingly want integrated solutions rather



than managing multiple vendors across the powertrain,” industry experts told *businessline*.

Traditionally, automakers sourced drivetrain, transmission and thermal-management components from different suppliers and integrated them in-house. The combined Dana-Eaton entity

will be able to offer a wider range of powertrain solutions under a single umbrella.

The implications could be particularly significant for India's commercial vehicle industry, where electrification remains at a relatively early stage. “The toughest EV challenge in India is not passenger cars. It is buses, trucks and light commercial vehicles that operate in extreme temperatures, congested traffic and demanding load conditions,” experts said.

They noted that Dana's expertise in e-axes and thermal management complements Eaton's strengths in transmission and electrification technologies. Bringing those capabilities together could help vehicle

manufacturers develop electric commercial vehicles better suited to Indian operating conditions.

LOCALISATION FOCUS Industry observers also see the merger strengthening India's role within global automotive supply chains. Both companies already have engineering and manufacturing operations in India, and experts believe the combined entity could further expand local sourcing and development activities as it seeks to achieve operational efficiencies and cost synergies.

“India sits at the intersection of the three biggest themes shaping the global auto industry today — electrification, localisation and cost competitiveness. Any

supplier looking for scale in those areas will inevitably view India as a strategic market,” industry experts told *businessline*.

Some experts also see an opportunity for India to deepen its position as an export and engineering hub as global suppliers optimise manufacturing footprints.

However, experts cautioned that the consolidation could alter competitive dynamics within the supplier ecosystem. “The benefits of integration are clear, but consolidation also changes bargaining power. OEMs may gain engineering efficiencies, but they could face a supplier with greater scale and stronger negotiating leverage, particularly in commercial vehicle powertrains,” they said.

Indian Army drops colonial-era dress customs

The changes are elaborated in the newly issued Army Uniforms-2026 Pamphlet

Asian News International
New Delhi

The Indian Army has removed several colonial-era vestiges from its dress regulations, the mandatory carriage of swords by reviewing officers, and the use of pouch belts with certain mess dresses. It has also dropped the use of archaic terminology such as “Royal”.

The Army is introducing the indigenous Bandi jacket as part of civil formal dress in a broader effort to align military traditions with India's sovereign identity.

The changes are elaborated in the newly issued Army Uniforms-2026 Pamphlet, which standardises dress regulations across the force while introducing what the Army describes as a deliberate move



NEW ATTIRE. Apart from the symbolic changes, the Army has introduced a new winter working dress featuring a battle jacket, which will replace the existing jersey-based uniform by June 2029

away from residual colonial-era symbols and terminology.

The rationale behind the changes is explicitly outlined in a chapter titled “Indigenisation and Alignment with National Ethos.” The

manual states, “In keeping with the nation's sentiments and evolving sovereign identity, a number of deliberate refinements have been incorporated in this edition of Army Uniforms Pamphlet.”

INDIANISATION PUSH Among the changes is the introduction of a closed-neck coat, commonly known as a Bandi jacket, to be worn over a full-sleeved shirt with matching formal trousers and closed footwear, bringing an indigenous element as part of formal civil attire for officers. The manual authorises that the Army has also removed the pouch belt from Mess Dress Nos ‘S’ and ‘G’, made the carriage of swords by Reviewing Officers optional, and discontinued the use of archaic terminology such as “Royal”, all of which had remained embedded in military traditions inherited from the colonial period.

According to the manual, these measures are part of a wider review of legacy practices.

In another major change, the manual narrows the occasions on which swords may be carried.

Swords will now be carried only by parade commanders, contingent commanders and designated personnel during major ceremonial events such as Republic Day, Independence Day, Army Day parades and Guards of Honour.

Apart from the symbolic changes, the Army has introduced a new winter working dress featuring a battle jacket, which will gradually replace the existing jersey-based winter uniform (Dress 3A) by June 2029.

It bars radical hairstyles, unauthorised beards, visible electronic gadgets, tattoos, body piercings and cosmetic make-up while in uniform.

20 rebel TMC MPs to merge with Tripura’s Nationalist Citizens Party

Our Bureau
Kolkata

In the latest jolt for Trinamool Congress chief Mamata Banerjee, a rebel faction comprising 20 party MPs on Sunday decided to merge with the Tripura-based Nationalist Citizens Party (NCP), a little-known Bengali-centric political outfit. The rebel MPs met Lok Sabha Speaker Om Birla and formally communicated their decision. They also sought separate seating arrangements in Parliament.

Speaking to reporters after meeting Birla in New Delhi, Kakoli Ghosh Dastidar, who is leading the rebel faction, said: “We (20 MPs) are merging with the Nationalist Citizens Party. We have requested separate seating arrangements in Parliament.” A four-term parliamentarian, Dastidar claimed that

the group enjoys the support of more than two-thirds of the TMC's Lok Sabha MPs.

“We command a two-thirds majority. We will work for the country and support the NDA under the leadership of the Prime Minister,” she said.

The development comes amid a deepening crisis within the Trinamool Congress following its crushing defeat to the BJP in the recent West Bengal Assembly elections. The party currently has 28 MPs in the Lok Sabha.

Earlier in the day, TMC MPs Kirti Azad and Sagarika Ghose, aligned with the Mamata Banerjee camp, met Birla and submitted a letter from party general secretary Abhishek Banerjee.

The letter urged the Speaker not to recognise any separate bloc or faction claiming to represent the Trinamool Congress. Meanwhile, the rebellion

within the party's legislative wing has also gathered momentum. Last Wednesday, Ritabrata Banerjee, leader of the rebel faction in the West Bengal Assembly, claimed the support of 64 MLAs, up from 58 earlier.

“The count is now 64 MLAs. These legislators will submit a letter to the Speaker. We are the real Trinamool Congress,” Banerjee told reporters outside the Assembly.

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QUICKLY.

Jio Platforms tops India in global patent filings

Mumbai: Reliance Industries' Jio Platforms ranked in the global top 20 in the Patent Co-operation Treaty (PCT) rankings for 2026, released by the World Intellectual Property Organisation. This places Jio among an elite group of global technology leaders like Huawei, Qualcomm, Google and Apple, and establishes India-origin innovation in the global technology creation landscape. Jio Platforms has filed 6,817 patents cumulatively as of March 31. **OUR BUREAU**

Two urea plants to start production soon

New Delhi: The government highlighted that two new urea plants will soon start production with a total annual capacity of 25.4 lakh tonnes, a development that would help the country reduce its import dependence. India imported more than 100 lakh tonnes of urea during the 2025-26 fiscal year. The Fertilizer Ministry highlighted the achievements of the Modi government in this sector. **PTI**

In France, Modi says that the world is awaiting human-centric technologies

PUSH FOR CHANGE. The Prime Minister also highlighted his government's reform initiatives in several sectors

Press Trust of India
Nice (France)

India is no longer just a consumer of global solutions, but a major contributor to them, Prime Minister Narendra Modi said on Sunday while laying out New Delhi's vision for developing human-centric innovation and technologies.

The Prime Minister made the comments at the inauguration of the 'Bharat Innovates' event in presence of French President Emmanuel Macron in this Mediterranean French city.

"India has demonstrated that innovation and inclusion are not contradictory but complementary. This means that the greatness of any innovation lies not merely in its evaluation, but in its human impact," he said at the event, which brought together top innovation start-ups from India, France and other countries. "Until a decade ago, the



TECH INNOVATION. PM Narendra Modi addressing the gathering during the inauguration of the 'Bharat Innovates' event in Nice, France, on Sunday **PTI**

world viewed India as a nation that adopted technology. Today, India is emerging as a technology provider," Modi said.

BHARAT INNOVATES "Moreover, the innovations India creates and the solutions it offers benefit a vast segment of humanity. This is precisely the goal of Bharat Innovates," he said.

The Prime Minister also underlined the importance of technologies that are inclusive and human-centric. "Today, the world is looking towards technologies that are reliable, inclusive, human-centric and aimed at the global good. In such times, India's priority is technology for humanity, human-centric innovation," Modi said.

Cognizant's India listing plans go slow amid market volatility and AI impact

Sindhu Hariharan
Janaki Krishnan
Chennai/Mumbai

Cognizant has made little progress on its plans for an India listing, 10 months after the proposal was first publicly discussed, according to sources familiar with the matter.

Sources told *businessline* that the listing plan had remained in the preliminary, exploratory stage, and is taking longer than earlier anticipated, given the performance of the India markets

and the AI impact on the IT services industry. The company has yet to appoint investment bankers.

In October last, the Nifty50 was above 25,800 levels. Since then it has fallen nearly 9 per cent. Further, Cognizant's own shares, listed on the Nasdaq, have fallen around 39 per cent in the last six months.

Cognizant first spoke about its India listing in October 2025, while announcing its July-September quarter earnings, that it was assessing a potential primary offering and secondary list-

ing in India with its legal and financial advisers, and that the review was still in early stages. Ten months later, sources said that little progress had been made.

"Considering its stock performance and the uncertainty in the Indian markets, it will be difficult for the company to get a decent valuation," said an investment banker, who has worked on IPOs of high profile multinational companies.

The biggest blow to the company's stock is its recent ouster from the Nasdaq-100



Index after more than a two-decade run.

AI WAVE Nasdaq on Thursday announced that Cognizant will be removed from its Nasdaq-100 Index this month as the index makes room for a set of new companies, many of which are riding the AI

He also highlighted his government's reform initiatives in several key sectors, including high-technology, nuclear energy, defence and innovation.

"This decade is one of both disruption and development for the world. Amidst conflicts and the escalating impact of climate change, the world is undergoing a period of unprecedented upheaval," he said.

CHALLENGES FACED "Yet, while humanity faces numerous challenges today, opportunities also abound. Artificial intelligence, quantum computing, biotechnology, space technologies, and advanced materials are the technologies that will shape the future of humanity. These represent the next chapter of human civilization," he said.

Modi said every technological revolution offers humanity a new opportunity, and every opportunity brings with it a new responsibility. "Bharat Innovates is an in-

itation to the world to co-create the next chapter of global innovation with India," the PM, who landed in Nice on Saturday night, said.

Modi further said India is innovating for a sustainable future. "India is innovating for the world," he said, adding that innovation is in the country's DNA.

"India has emerged not as a consumer of solutions but as a contributor to solutions in the world," he noted. The Prime Minister also spoke about the strength of India-France relations.

In his address at the event, President Macron said France respects the Make-in-India initiative and that Paris has been part of it in diverse sectors. Describing India as a "country of innovation," he said New Delhi and Paris had a "true partnership" in critical sectors like AI and climate change.

There is scope to expand bilateral cooperation in the civil nuclear energy sector, including in the area of small modular reactors, he said.

wave. "It will be difficult to convince investors," said the banker, pointing out that several large companies were delaying their IPOs due to valuation mismatch.

LISTING PLANS However, Cognizant denied that the listing plans had been shelved.

"As we mentioned at the end of April, we continue to make progress and advance on our evaluation of potential primary offering and secondary listing in India. We remain committed to acting in the best interest of our

shareholders. That process is ongoing," a spokesperson said in a statement.

In April, while announcing its Q1 earnings, Cognizant also signalled layoffs with the announcement of a transformation programme, called Project Leap.

Under this, Cognizant expects to incur \$200 million-\$270 million of employee severance and personnel-related costs in 2026. Cognizant CEO Ravi Kumar S had said then that its AI builder strategy and partnership ecosystem positions it to bridge the AI gap.

Sharp fall in gold prices to exert pressure on jewellers' margins

Suresh P Iyengar
Mumbai

The sharp fall in gold prices, coupled with subdued demand, is expected to exert pressure on margins of jewellery companies in the June quarter.

Gold prices have been on a steady fall both in the domestic and international markets despite the simmering war between the US and Israel. The yellow metal has fallen by ₹25,195 or 14 per cent to ₹1,44,782 per 10 g from its high of ₹1,69,977 per 10 g recorded on May 13, in line with the global market trends.

This fall has occurred despite the import duty being increased to 15 per cent from 6 per cent in May to curb imports.

In the US, spot gold slipped to its lowest level in nearly 11 weeks to \$4,181 per ounce on Thursday. While a fall in prices usually leads to higher demand, gold jewellery buyers tend to post-

pone purchases in anticipation of further declines.

The margins of leading jewellery companies have already come under pressure in Q4 and retailers carrying inventory bought at higher prices could potentially face mark-to-market losses.

HIGHER IMPORTS Gold imports surged 24 per cent to \$72 billion in FY26, prompting Prime Minister Narendra Modi to appeal to consumers to reduce purchases in order to limit the impact on the current account deficit.

Anil R, Senior Research Analyst, Geojit Investments, said, "The fall in gold prices from March highs is expected to weigh on June quarter margins for jewellery companies due to high-cost inventory and elevated stocking levels in Q4."

He added that while improvements in product mix, making charges and hedging strategies may provide some cushion, they are unlikely to fully offset the impact of the

sharp price decline.

Sunil Katke, Head of Commodities Retail Business, Kotak Securities, said, "Jewellers may attempt to lower the impact of the fall in gold prices by increasing making charges and adopting fixed making charges per gram rather than linking it to gold prices."

He added that most large organised jewellers also maintain hedging strategies on the MCX to limit the impact of gold price volatility.

"We do not expect a meaningful slowdown in jewellery demand solely due to the Prime Minister's remarks. The comments appear to be aimed at reducing India's dependence on imported gold rather than directly discouraging domestic jewellery purchases," he said.

With the prices declining from their highs, retailers who bought gold at elevated levels may incur some inventory-related losses, which could affect margins during the quarter, Katke added.

Banks may face a tricky situation while seeking FCNR (B) deposits

K Ram Kumar
Mumbai

Banks may face a tricky situation in their bid to attract fresh Foreign Currency Non-Resident (Bank) deposits under the RBI's recent measures to attract foreign capital.

Depositors with existing FCNR (B) deposits denominated in US dollars may prematurely close them and roll-over the proceeds into fresh deposits to benefit from higher interest rates that the banks are currently offering for a limited period, according to bankers.

INTEREST RATES Banks sharply increased interest rates on FCNR (B) USD deposits to 6-7 per cent from the earlier 3 per cent-odd levels.

This comes in the wake of the RBI's announcement, as part of its June 5 measures, to attract dollars to stabilise the rupee. The RBI announced that it will bear the



full hedging cost on fresh 3-5-year FCNR (B) deposits that banks mobilise up to September 30, 2026.

Moreover, such deposits have been exempted from statutory pre-emptions such as the cash reserve ratio and statutory liquidity ratio. Within the overall non-resident deposits of banks, the outstanding under FCNR (B) deposits category nudged up about 3 per cent year-on-year to stand at \$33.756 billion as of March-end 2026 against \$32.809 billion as at March-end 2025, as per RBI data.

Inflows into these depos-

its were sharply lower at \$946 million in FY26 against \$7,076 billion in FY25.

"About 30 per cent of the existing FCNR (B) deposits may be prematurely closed and rolled over into fresh deposits. It is unclear if RBI will bear the hedging cost of such rolled over deposits. If it does not, then banks will have to pick up the tab. Banks simply cannot afford to lose existing depositors to competition," said a senior private sector banker.

PREMATURE CLOSURE "In case a deposit is sought to be prematurely closed after it has run beyond the one-year period, banks usually pay the interest rate applicable for the period the deposit has run (instead of the contracted rate). There could also be a penalty (of up to one per cent in the case of some banks)," banking expert V Viswanathan said.

SBI economists expect around \$40-45 billion to come in through the FCNR (B) deposit route.

Nitin Gadkari clears E100 fuel framework

Amit Vijay Mohile
Mumbai



Union Minister for Road Transport and Highways Nitin Gadkari speaking at the conference on Saturday

India has formally approved the regulatory framework for E100 fuel, allowing vehicles to run on 100 per cent ethanol for the first time. Union Minister for Road Transport and Highways Nitin Gadkari said he had signed the regulations on Saturday, a move that could accelerate the rollout of ethanol-powered cars and two-wheelers, create a new market for flex-fuel vehicles and help reduce the country's dependence on imported crude oil.

Speaking at the Sugar, Ethanol & Bio-Energy India Conference in Nagpur on Saturday, Gadkari said the approval creates a legal framework for automakers, fuel retailers and testing agencies to begin commercial deployment of pure ethanol-powered vehicles.

"I signed the file at 8 pm today (Saturday)," Gadkari said, adding that the decision would help reduce dependence on imported fossil fuels while expanding the use of domestically produced biofuels.

The move takes India beyond its ethanol-blending programme, which is nearing nationwide implementation of E20 petrol.

By creating a regulatory pathway for E100, the government is enabling vehicles designed to run entirely on ethanol, adding another technology option alongside electric, CNG, hybrid and hydrogen-powered mobility.

FLEX-FUEL VEHICLES Gadkari said flex-fuel vehicles are already in the market. "Maruti Suzuki has launched the WagonR flex-fuel vehicle, and Hero has introduced motorcycles that can run on 100 per cent ethanol," he said. "Toyota and Hyundai are also expected to launch vehicles that can run on 100 per cent ethanol."

The regulatory clearance gives manufacturers certainty to accelerate investments in flex-fuel technologies. E100-compatible vehicles require specialised engine calibration and fuel-system components to handle ethanol's distinct properties.

For automakers, the move creates an additional decarbonisation pathway at a time when the industry is investing across multiple power-train technologies.

The government also sees ethanol as a tool to support rural incomes. Gadkari described the sugar and bio-energy sector as "the backbone of rural development" and said higher ethanol production could create additional demand for sugarcane, maize and other feedstock used in biofuel production.

The policy is expected to support India's efforts to reduce its dependence on imported crude oil.

NEW FRAMEWORK The country imports more than 85 per cent of its crude requirement, making fuel one of the largest contributors to the import bill.

By increasing the use of domestically produced ethanol, the government hopes to improve energy security while reducing exposure to volatile global crude prices.

While the regulatory approval removes a key hurdle, largescale adoption will depend on fuel availability and infrastructure.

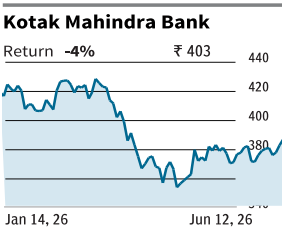
Oil marketing companies are expected to gradually roll out E100 dispensing stations, while vehicle-certification and fuel-distribution systems will need to align with the new framework.

The approval transforms ethanol from a blending component into a standalone transport fuel, potentially paving the way for India's first largescale market for ethanol-only vehicles.

TODAY'S PICK.

Kotak Mahindra Bank (₹403.35): BUY

Akhil Nallamuthu
bl. research bureau



The stock of Kotak Mahindra Bank has been rallying since early April as it found support at ₹350. Last week, it extended the uptrend and posted a gain of nearly 7 per cent.

Consequently, it made a higher high and also broke out of resistance levels at ₹393 and ₹400, opening the door for further rally. While there is a chance for the stock to retest the resistance-turned-support of ₹393 and ₹400, we expect it to resume the uptrend and eventually appreciate to ₹430.

Therefore, traders can buy the stock now at ₹403 and accumulate at ₹395. Place the initial stop-loss at ₹382. When the stock rises to ₹416, revise the stop-loss to ₹405.

On a rally to ₹423, tighten the stop-loss further to ₹416.

Book profits at ₹430.

Note: The recommendations are based on technical analysis. There is a risk of loss in trading

businessline.

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Blooming paradox: Flower industry soars even as North-East withers

Government started collecting data on the area and production of flowers only from 1993-94

Prabhudatta Mishra
New Delhi

India's floriculture sector is in full bloom. The area under flower cultivation has touched a record 4 lakh hectares in 2025-26, recovering sharply from a low of 2.82 lakh hectares in 2021-22.

Gross value added from the sector has surged from ₹39,287 crore in 2019-20 to ₹51,643 crore in 2023-24.

Yet, one just has to travel to the mist-covered hills of Arunachal Pradesh, and a very different story unfolds.

In 2009-10, the State cultivated 1,200 hectares of cut flowers, producing 2,860 tonnes annually, enough to rank it among the country's top two anthurium produ-

cers, trailing behind only Mizoram.

Today, that bloom has all but vanished.

Farmers have drifted to other crops, and the anthurium, once the pride of Arunachal Pradesh's hills, has migrated south and east, finding new enthusiasts in Meghalaya, Nagaland, Bihar, West Bengal and Kerala.

OTHER CROPS

The national boom is being driven not by new frontiers opening up but by old ones holding firm; farmers in established growing States are expanding acreage despite mounting pressure from urbanisation and surging real estate values, racing to extract more from the land

Flower production in 2025-26*

	Area in '000 hectare	Production in '000 tonnes	
		Loose	Cut
Marigold	106.19	1,277.35	13.71
Rose	48.20	218.23	620.45
Chrysanthemum	41.19	962.78	42.80
Jasmine	31.61	322.10	0
Tube Rose	27.19	239.67	111.88
Gладиолус	11.78	10.74	177.19
Gerbera	5.66	9.742	6.76
Anthurium	3.09	0	8.97
Orchids	2.42	22.09	7.28
Carnation	2.24	0.13	15.49
Other Flowers	119.93	318.05	178.68
Total	399.52	3,380.87	1,203.22

*2nd Advance Estimate; Source: Agriculture Ministry

before it is swallowed by the ever spreading cities.

In the North-East, by contrast, flower cultivation is losing ground to more im-

mediately lucrative crops, hampered further by infrastructure gaps and the absence of reliable market linkages.

Sikkim, a leading grower of orchids, reported the total area under flower cultivation at 242 hectares and production of 16,509 tonnes in 2025-26, which has been maintained at the same level as a decade ago.

AREA & OUTPUT

In 2015-16 also, the area and output were the same.

On the other hand, the area under flowers in Mizoram has shrunk to 50 hectares in 2025-26 from 240 hectares in 2017-18.

Nagaland has done exceptionally well as the area under floriculture reached 1,190 hectares in 2023-24 from only 70 hectares in 2015-16, official data show. India started collecting

area, production and other data on flowers only from 1993-94.

The share of floriculture in overall horticulture production, which reached a record 377.78 million tonnes and higher than even food-grain output, has jumped from 0.76 per cent in 2014-15 to 1.2 per cent in 2025-26.

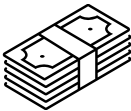
TOP STATES

Among the top States in flower production during 2023-24, the highest share was recorded by Tamil Nadu (19 per cent), followed by Madhya Pradesh (14 per cent), Karnataka (12 per cent), Andhra Pradesh (11 per cent) and West Bengal (10 per cent).

Hanging in balance

Capital flows may just edge up to offset higher CAD

India’s current account surplus of \$7.1 billion in Q4 of FY26 comes as a pleasant surprise — given the spike in crude oil prices. While the current account balance has been favourable in Q4 over the last four years, it is notable that the Q4 surplus was recorded in difficult times. Both US tariffs and the Iran war (which began in February) were factors. The surplus came about, thanks to an increase in services exports — led by computer and other business services — and, rather surprisingly, strong overseas remittances.



However, FY26 as a whole recorded a marginally higher current account deficit (CAD) of \$25.2 billion, or 0.6 per cent of GDP (\$23.3 billion in FY25). This increase was largely on account of the expansion in goods trade deficit by \$51 billion. While US tariffs and their effects contributed to a higher goods trade deficit in FY26, rising oil prices have made matters worse this fiscal. Analysts have estimated the CAD for FY27 to be above 2 per cent of GDP. This looks plausible for various reasons. First, higher energy prices will lead to a spike in the oil import bill, unless the war ends in West Asia soon. Second, the CAD also depends on the extent to which remittances can offset a higher goods trade deficit. Finally, it is to be hoped that the newly minted FTAs, such as those with the EU and UK, do not result in higher net imports. Ordinarily, a CAD of above 2 per cent of GDP is not threatening. India has financed higher CADs with capital flows. However, the problem is that FPI equity outflows, at above ₹2.5 lakh crore so far in 2026, have overshoot ₹1.6 lakh crore for all of 2025.

Debt flows, too, have been negative. Net NRI deposit inflows in FY26 were down 11 per cent at \$14.4 billion. Similarly, external commercial borrowings had also ebbed 23 per cent to \$14.2 billion. The recent monetary policy announced a US dollar-rupee swap facility to banks on fresh FCNR(B) deposits of three-to-five year maturity. Th central bank has also agreed to take on the hedging costs on these deposits. With most banks hiking the rates offered on FCNR(B) deposits by 200 to 400 basis points, the hope is that inflows from this route could improve. A similar swap facility is also being offered to public sector undertakings to raise ECBs from overseas. With borrowing rates likely to be lower due to the swap arrangements, PSUs could raise more ECBs until December.

Equity outflows are driven by global factors. The RBI and Centre have, therefore, rightly targeted FPIs investing in debt. FPI inflows into Indian debt through the fully accessible route has been positive in 2025 and 2026, indicating demand from global bond funds. The removal of capital gains tax on G-Secs as well as exemption of income tax on interest earnings on these bonds will help. Passive inflows through global emerging market funds can improve going ahead. FY27 may yet turn out to be a reasonably normal year on the external account front.

OTHER VOICES.

The Guardian

Words do not stop when the bombing begins

Last week, thousands of readers gathered for a literary festival in Kyiv, risking air raids to hear from writers. Four brutal years of war have not destroyed the appetite for writing, but fuelled it. Russia’s extensive and systematic attempts to destroy Ukrainian culture, and therefore identity, have rightly received widespread attention. Over 700 libraries were damaged or destroyed outright within the first three years of the full-scale invasion. But that campaign has also spurred efforts to move away from Russian literature and the Russian-language titles that previously dominated the market. Ukrainian literature and publishing has flourished far beyond the powerful documentary accounts of war often awarded attention outside the country, with growing room for experimentation. (LONDON, JUNE 12)



Japanese businesses should speak out to the Takaichi govt

The Japanese Chamber of Commerce and Industry in China (JCCIC) has released the White Paper on the Chinese Economy and Japanese Enterprises. The paper acknowledges that China provides “significant market opportunities” for foreign companies while also recognizing various concerning trends in China-Japan relations. It outlines three hopes: political and diplomatic issues between the two countries will not be allowed to spill over into corporate activities or cultural activities; high-level dialogue and exchanges between the governments of both countries will be held regularly and reliably, regardless of international and political circumstances; the Chinese government and relevant authorities maintain and strengthen dialogue with Japanese companies under all circumstances. (BEIJING, JUNE 13)

LINE & LENGTH.



TCA SRINIVASA RAGHAVAN

Everyone thinks there is only one gorilla in the room, namely, AI. Actually there are two. The second is the absence of policy instruments to tackle the massive unemployment that awaits societies because of AI. But while everyone is hyperventilating over AI, there is pin drop silence over the absence of options before governments.

Meanwhile, the companies that stand to benefit from a widespread use of AI are peddling all sorts of justifications. The most intellectually grounded of these is the Jevons Paradox.

William Stanley Jevons was a 19th century British economist who said that the invention of the steam engine, which used coal more efficiently, would result in higher coal production and consumption, not less. That is, less coal per steam engine would lead to more engines being bought.

Microsoft’s Satya Nadella might not have heard of Jevons until recently but he used the paradox to say that AI would have a similar impact and that unemployment would not increase. Instead, it would increase. Others are recalling Joseph Schumpeter who said this sort of thing was inherent to capitalism and that it was good.

This may well happen, eventually, even though the IT boom of the 2000s didn’t lead to any great and permanent increases in productivity that later led to greater employment. There are several studies that show this.

However, even though the past need not always be an infallible guide to the future, we do need to ask: even if AI increases the efficiency of capital-use and that gives better returns to shareholders, what will it do to, and for, labour, especially in the medium term? If 10 programmers are replaced by one AI application, the cost saving will increase the dividends to shareholders by some amount but what of the 10 people who are now wondering how to pay their bills?

BYE BYE, MAYNARD

John Maynard Keynes, the most influential macroeconomist to have lived after Kautilya, said governments must spend money even if they didn’t have it to create demand that would, in

Will employment rise in age of AI?

The IT boom of the 2000s didn’t lead to any great and permanent increases in productivity that later led to greater employment



turn, create employment. But country after country gradually has forgotten that Keynes had a very specific situation in mind, namely, that of reviving deficient aggregate demand to utilise installed capacity.

As a result of this amnesia, led by the Left which has co-opted the religious Right, the Keynesian solution for reviving demand has gradually turned into massive welfare spending. This is not something that Keynes ever intended, although he is believed to have said that money should be put in people’s pockets even if it was just for digging holes in the ground.

To cut a long story short, because of

If 10 programmers are replaced by one AI application, the cost saving will increase the dividends to shareholders but what of the 10 people who are now wondering how to pay their bills?

this self-serving use of Keynes by politicians, we now have unsustainable levels of government debt in far too many countries. Debt now looms menacingly over the world.

Before 1968, government debt and unemployment were not linked. Now they are. For a while higher government debt did bring down unemployment. But not anymore. The goose that laid the golden egg for politicians has died.

The IMF has therefore been getting horror nightmares. It is mumbling away that there is no paddle to row out of this creek. But as I said above, no government is paying a blind bit of attention to its grumbling.

Finance ministers may occasionally warn the heads of their governments that their goose will be cooked but it’s not having much effect. Deficit financing to win some political advantage over rivals remains very much the go-to solution.

NO MONEY LEFT

The most devastating example of this came after the Atlantic financial crises of 2008 (as Rakesh Mohan, former deputy governor of the RBI, rightly calls it).

India’s ethanol push requires a relook

Higher ethanol blending targets will intensify water stress, without achieving energy security or a reduction in imports

Ritesh Kumar Singh

The logic behind India’s ethanol push appears compelling: blending more ethanol with petrol will help cut crude oil imports, ease pressure on the current account by conserving foreign exchange, support the rupee and raise farm incomes.

At a time when India imports the bulk of its crude oil requirements and external balances remain sensitive to oil prices, the argument has found wide acceptance. But the economics increasingly suggest otherwise.

The ethanol push now appears to have more to do with absorbing excess production capacity than reducing India’s import dependence. The challenge has shifted from building supply to finding demand.

India requires 10-11 billion litres of ethanol to meet E20 blending norms. Yet production capacity had already approached 20 billion litres (<https://tinyurl.com/2s3e6uza>) by November 2025 and may expand by a further 15 per cent in FY 2026-27 as new distillation projects come on stream. Put differently, the current blending target absorbs barely half of installed capacity. The issue therefore is no longer about producing more ethanol; it is about finding demand for capacity already built.

The rapid expansion of ethanol capacity was not accidental but actively supported through government-backed soft loans and interest subvention running into thousands of crores. The ethanol programme may have catalysed

investments exceeding ₹40,000 crore. Once investments of that scale are committed, utilisation becomes central to financial viability. Idle capacity compresses returns, raises fixed costs and increases the risk of stranded investments.

To address recurring sugar surpluses, cane-based distilleries were encouraged. As sugar availability tightened, policy support gradually shifted towards grain-based distilleries, including incentives through relatively higher prices for ethanol produced from maize. Ethanol created an additional revenue stream protected through administered pricing, unlike sugar, where prices are largely influenced by supply conditions.

In many ways, the current excess capacity reflects the cumulative effect of these policy shifts and incentives. Higher blending targets are no longer only an energy-security tool but are needed to absorb surplus capacity.

Substituting petrol with domestically produced ethanol does lower oil demand. But the import story becomes more complicated when the full production chain is considered.

Producing ethanol at scale requires larger volumes of sugarcane, maize and rice. Expanding these crops raises demand for fertilizers such as urea, DAP and potash which require imported inputs such as natural gas.

Greater policy support for ethanol feedstocks may also reshape cropping patterns, drawing acreage towards sugarcane, maize and rice and away from oilseeds or pulses — segments where India already relies heavily on imports. In that sense, ethanol may reduce



WATER USAGE. A major issue

crude oil imports without proportionately reducing external dependence. The dependence simply shifts. Lower oil imports could partly be offset by higher requirements for fertilizers, natural gas and edible oil imports, while consumers also experience lower mileage because of ethanol’s lower energy density.

ENVIRONMENTAL COSTS

The environmental arithmetic is equally complex. A government study estimates that producing a litre of ethanol from sugarcane requires roughly 2,860 litres of water across the production cycle. Water intensity rises further for grain-based pathways, with maize-based ethanol requiring around 4,093 litres and rice-based ethanol as much as 10,790 litres per litre of output.

India now has close to 500 ethanol distilleries (<https://tinyurl.com/eps3cdz4>), with grain-based capacity accounting for a growing share of incremental supply. The shift towards grain-based ethanol

Globally, governments have spent a few trillion dollars that they didn’t have in the name of social security and welfare.

Now even that option has vanished because it’s not going to help. It has become like firing blanks in the air: there’s a lot of noise and smoke but the problem doesn’t go away.

The world has over eight billion people now. Around half of them are of working age. But barely over a quarter of those four billion are employed on a full-time basis. The rest are self-employed earning basically subsistence wages.

Even without AI we are talking of at least two billion people who need income and other forms of support. This is Karl Marx’s “reserve army of the unemployed”. They constitute not only an economic challenge but a social and political one as well. In India, this situation will worsen quickly as we have already fallen below the population replacement rate. That is, India will almost certainly grow old before it gets rich.

To meet these challenges a lot of money is needed. Money that’s no longer available to governments.

● BELOW THE LINE



Jugaad!

To make computerisation more palatable to bank employees, trade unions and associations, the exercise was called “mechanisation” in the late 80s, according to SBI Chairman CS Setty. Narrating an interesting anecdote at the recent Citi Investors’ Conference, he noted that Indian banking then was dominated by public sector banks and the

government of India, through RBI, wanted to implement computerisation. “They (the government) didn’t call it computerisation. They called it mechanisation. They wanted to bring machines in the banking industry. Every employee, union, and association pushed back. They didn’t want machines,” Setty said. So, to get buy-in from all stakeholders, the government came up with a very typical Indian “jugaad” — anybody who uses a computer will get an increment.

Match fixing?

The Agricultural and Processed Food Products Export Development Authority (APEDA) recently called for applications to fill the post of

Director, Basmati Export Development Foundation. It asked interested applicants to visit its website for more details. Details on the site said applicants from the private sector should not be over 55 years of age, while retired bureaucrats should not be over 62. A little bird says this rules out a former bureaucrat who was among those short-listed initially and another person who had been thought of as a favourite to fill the post.

Trust factor

“I don’t know you. I am talking to you for the first time. How can I trust you? How do I know that you are what you claim you are over telephone?” These questions came

from the chairman of a PSU when contacted for a routine media query. Even after this reporter introduced himself, explained the story and pointed to credentials available in the public domain, the executive remained unconvinced. “Is this telephone number listed in those credentials?” he asked. “If I had met you, I would speak to you. We cannot trust anyone in this fake age.” The two-minute exchange captures a growing reality of modern reporting: before journalists can verify facts, they increasingly find themselves having to verify their own identities. In a world of spoofed calls, fake social media profiles and AI-generated impersonations, even a straightforward request for

comment can begin with a trust deficit.

Checkmate

Tamilaga Vettri Kazhagam leader C Joseph Vijay may have outmanoeuvred political rivals in the recent Assembly elections, but he met his match on the chessboard on Monday. Fresh from winning the Norway Chess Championship, Grandmaster R Praggnanandhaa visited the Secretariat and engaged Vijay in a friendly game. The TVK leader made a few impressive moves, but the young chess champion quickly seized the initiative and secured victory. Vijay presented the Tamil Nadu grandmaster with a cash award of ₹50 lakh for winning the title at Norway. **Our Bureaus**

The writer is a business economist and CEO, Indonomics Consulting Private Ltd

ON THE CAMPAIGN
TRAIL.

Football mania is sweeping the world and the biggest global brands have upped their game with breathtaking commercials, full of ambition and emotional and cultural resonance.

Nike rips the script

Back in 2010, Nike scored sensationally with its *Write the Future* World Cup commercial, as it captured how one pass or tackle can change the destinies of players and nations. The 2026 *Rip the Script* ad is set off the pitch and brings a different energy — raw, chaotic vibe. Created by Wieden+Kennedy, the ad begins on a film set where football megastars, irritated over delays and an overbearing director, rebel and race off — colliding with entertainment world’s superstars. The starry ensemble features footballers Kylian Mbappé, Cristiano Ronaldo, Erling Haaland, Vini Jr, Zlatan Ibrahimovic, Ronaldinho and Wayne Rooney, and sports and cultural icons LeBron James, Travis Scott, Kim Kardashian, Channing Tatum and Ted Lasso. It’s unscripted fun, gut play and fearless talent blended with humour.

Backyard stories

adidas delivers a kick in its ambitious World Cup film *Backyard Legends*, where actor Timothée Chalamet joins Lionel Messi, Bad Bunny, Lamine Yamal, Jude Bellingham, Trinity Rodman, Zinedine Zidane, David Beckham and Alessandro Del Piero to tell a story of neighbourhood lore. The film celebrates the joy of free play and sets out to remind fans that every pitch, from backyards to the world’s biggest stage, can create a legend. Set against a nostalgic soundtrack, 1990s street and terrace style, analogue tech and era-defining hairstyles, the film blends CGI and visual effects to bring to life a simple truth: The backyard is where pressure disappears and legends begin. Official Match Ball provider and kit supplier to 14 federations competing at the FIFA World Cup, adidas is also showing up in backyards with a lot of activations. Truly on the ball.

Piecing it together

Lego’s World Cup 2026 campaign, *Everyone Wants a Piece*, has justifiably gone viral. It features animated minifigures of soccer legends Lionel Messi, Cristiano Ronaldo, Kylian Mbappé and Vinicius Júnior in a dimly lit room, all competing to assemble and claim a golden Lego World Cup trophy. The competition is intense and the viewer gets highly involved. But in a playful twist, before any of them can finish, a young child enters the frame and completes the task, putting his own Lego figure on top.

Bubbling up

Coca-Cola’s World Cup campaign *Feel It All* focuses on football fandom and the rollercoaster emotions of supporters. The latest spot film in the campaign, *No Better Feeling*, developed by WPP OpenX and led by Ogilvy, acts as a visual love letter to football culture by zeroing in on the sport’s most tension-soaked moments — the video assistant referee (VAR) check. Featuring José Mourinho and J Balvin and commentary from Peter Drury and Luis Omar Tapia, the campaign captures the emotions of fans all over the world as they wait to see if it’s a late-game goal or not. The intense match electricity and the agony and ecstasy of fans are beautifully portrayed.

AGENCY WATCH

‘We’re increasingly acting as consultants, rather than just advertisers’

Chitra Narayanan

Innocean was born in 2005 in South Korea as a marketing and communications company of the Hyundai Motor Group, but today the integrated agency with a presence in 23 countries also has a growing portfolio of clients other than the auto major. Santosh Kumar, Chief Operating Officer of Innocean India, headquartered in Gurugram, says 30 per cent of revenues in the country now come from non-Hyundai, non-Kia work.

How has Innocean India’s portfolio evolved beyond the Hyundai group? The origin and genesis of Innocean was built around our affiliated clients, Hyundai and Kia. Over the last decade-plus, the focus was on streamlining operations and building the complete ecosystem for those clients. But a key agenda for me has been to put

Innocean India on a growth pedestal — and that requires both vertical and horizontal expansion. Within Hyundai and Kia, we’ve gone deeper into services beyond just creative and media mandates. But the larger growth in the last three years has come from non-affiliated clients — operating as an independent agency in the market. We work with more than 15 clients, and 30 per cent of our revenue comes from non-Hyundai/Kia work.

Who are among your non-affiliated clients? We work across multiple verticals. We have Narayana Healthcare. We handle Sony. In energy, we work with Valvoline and Suzlon. In real estate, DLF is a large client. In lifestyle and FMCG, we have Swiss Beauty and Kajaria Ceramics. None of our non-affiliated clients are Korean — we’re targeting mid-sized Indian brands and regional players, where we can make a difference.

Some agencies say customer

relationship management (CRM) and data services account for 50 per cent of business. Is that so at Innocean too? A significant portion of revenue is coming from these streams, and we clearly see that in two to three years they will contribute substantially to our revenue ecosystem. Every agency has to move in this direction — it’s not a choice, it’s what clients need. We’re increasingly acting as consultants, rather than just advertisers. We’ve built a centre of excellence around data, martech (marketing technology) and CRM, and those capabilities are generating stronger client attraction.

You opened an office in Bengaluru recently. Is it your first expansion outside Delhi NCR? Bengaluru is our first domestic expansion. We already have a team in place, and one client — Narayana Healthcare. Two more clients are being acquired. The branch head will be finalised in

K-MARKETING CRED. Santosh Kumar, COO, Innocean India

about a month. Mumbai is next on the roadmap. We’re building Bengaluru as a native AI-led setup. Start-ups there are early adopters of technology, have large ambitions, and need the agility and flexibility we offer. It creates a synergistic environment to trial new AI solutions, including multi-agent models.

The industry is going through turbulence and consolidation. Does having Hyundai and Kia

as anchor clients protect you? There has been high turbulence in the past year due to changing demands, technology emergence, and shifting consumer behaviour. But two things give Innocean an advantage. First, we’re not small enough to struggle with cash flows or working capital, but we’re also not a large bureaucratic network agency — so we bring genuine agility and flexibility to clients. Our Korean DNA, which is very process-oriented, actually helps us build lean, productive structures. Second, when most agencies struggled to grow even in single digits last year, we’ve grown in double digits every year for the last three — roughly twice the advertising industry’s growth rate.

How much has agentic AI seeped into advertising? Honestly, no agency, including us, has made a holistic shift to agentic AI yet. Advertising runs on creativity, which is fundamentally human. At Innocean, we champion human-centric AI: AI

handles automation, speed and scale, while creativity, strategic thinking and emotional storytelling remain human-led. That said, in areas suited to automation — data analytics, research and trend-finding — we have fully deployed agentic AI.

Unilever announced moving a large chunk of advertising budget toward influencers and digital. Is that trend playing out with your clients? Digital advertising has already crossed 50 per cent of ad spend by most reports. Creator economy and influencer marketing are part of that. But the role varies by category. In FMCG, lifestyle and fashion, creator adoption is very high — it’s an aspirational, visually-driven decision. In automotive, the creator economy is growing but still requires care: We need influencers who create brand impact, not just reach and engagement metrics. Quality of influence matters more than scale.

What are the big trends in

advertising you’re seeing? The industry is going through a transition it has never seen before. The key trends I see:

- AI will rewrite the industry playbook. Agencies that embrace it will augment productivity and competitiveness; those that remain in denial will fall behind.
- Livestreaming and sports marketing will continue to grow. India is expanding beyond cricket, and brands are following. Innocean has already partnered for ICC campaigns, and we’re investing more in sports marketing.
- Regional content is on the rise, driven by OTT. This also accelerates the CTV (connected TV) trend.
- Micro-dramas are an emerging and genuinely exciting content format.
- The measurement model is changing. It’s no longer just about reach, frequency and share of voice — it’s about share of engagement, relevance and brand resonance. Metrics are evolving to match.

HERITAGE TRAIL

Celebrating the iconic Bengali brands

90 NOT OUT. A brand conclave and a book spotlight the region’s enduring entrepreneurial legacy — from biscuits to paints and clubs

Mithun Dasgupta

In 1892, a small biscuit venture began in Calcutta, now Kolkata, with an investment of ₹295. The founders bet on the city’s growing appetite, fuelled by booming trade and an expanding middle class. That bet paid off. Over the next few decades, the company, Britannia, revolutionised India’s snacking sector, being the first firm east of the Suez Canal to install industrial gas ovens. Later, it pioneered high-quality sliced and wrapped bread.

Britannia Industries’ turnover crossed ₹18,000 crore in the last financial year. What’s more, the born-in-Bengal biscuit and bread maker’s products are today exported to over 80 countries. And the FMCG major continues to build strong consumer connection through a portfolio of bold, creative and culturally resonant campaigns. Britannia’s story features in *Brands of Bengal: 90 Not Out*, a compendium that chronicles brands that originated in the State and are older than nine decades. Another gripping story in the book is that of Dabur. The year 1884, eight years before brand Britannia began its journey, saw ayurveda embrace modernity for the first time, in the bylanes of Calcutta. “Dr SK Burman was deeply concerned about how inaccessible medicine was in rural Bengal, and was gripped by a desire to provide affordable cures to people living in remote villages. Thus, he started preparing ayurvedic remedies for malaria and cholera, and cycling to



ORIGIN STORIES. (Left) Shibasish Bandyopadhyay, research scholar, and Diptangshu Mondal, proprietor, Deep Prakashan, unveil *Brands of Bengal* at Brand Charcha in Kolkata; (right) born-in-Bengal Britannia biscuit goes places



deliver them to people’s homes. Dabur, an abbreviation of ‘Daktar Burman’, became the name for this mission,” recounts the chapter on the company.

TIMELESS TALES The book captures the spirit of Bengal’s entrepreneurial legacy and timeless brands. Apart from Britannia and Dabur, there are stories of time-honoured firms like ITC, CESC, Berger Paints, Saregama, Eveready Industries, Baidyanath, Luxmi Group, Shalimar Paints, Boroline and Cookme. It also traces the brand journey of legendary restaurants and bakeries like Trincas, Aminia, Mitra Cafe, Flurys and Nahoum and Sons; elite and heritage clubs such as Calcutta Club, The Bengal Club, Calcutta Rowing Club, Calcutta South Club, Royal Calcutta Golf Club, Royal Calcutta Turf Club and Tollygunge Club; and celebrated football clubs like East Bengal, Mo-

hun Bagan and Mohammedan Sporting. No football match in Bengal is complete without sweets. Fittingly, the book unpacks the journeys of timeless sweet brands — KC Das, Balamullick & Radharaman Mullick, Bhim Chandra Nag, Girish Chandra Dey & Nakur Chandra Nandy, and Sen Mahasay. The book was launched at Brand Charcha, a first-of-its-kind business and cultural conclave held in Kolkata earlier this month with the idea of championing the State’s rich commercial history and spotlighting emerging regional enterprises. Rather fitting when you think how regional narratives are growing stronger. Recently in the South, for instance, SuperChennai has been trying to celebrate the Tamil Nadu capital’s legacy and position it on the global stage. Brand Charcha — put together by Radium Management, an integrated communication and creative

agency — brought together storytellers and entrepreneurs, and the event saw a lot of discussion on Bengal’s generational brand wealth. The book was the showpiece. “There is no readily available handbook or compilation on legacy brands that emerged from Bengal or a brief history of business houses of Bengal. Where would students and enthusiasts of the business history of Bengal reach out to?” said Subhadip Choudhury, Co-founder and Director, Radium Management Services, adding this is the core of the initiative. According to Choudhary, this is but the start. “This is the first in the series on popular brands (products, institutions, associations) from Bengal that have been in existence for 90 years and above,” he said, adding there are many brands that did not make the list either because of the business period or that they did not make that much of an impact on business and society but

still deserve due mention. “Our next series would cover that with proper diligence.”

STAYING RELEVANT What are the factors that ensure a brand’s longevity? “Endurance is less about being remembered and more about being believed, again and again. A brand stays relevant when its behaviour keeps matching its promise, so trust quietly compounds,” said Samrat Mukherjee, Vice President, Madison World.

“Boroline has stayed a part of Bengali life for almost a century with barely any advertising, because it kept a small promise faithfully and became a daily habit. Reliance isn’t won in the ad; it’s won in what happens after the purchase. A brand people find easy to believe rarely has to shout. Refresh how you express yourself as often as you like — just don’t change what you’re asking people to trust,” Mukherjee emphasised.

Currently, Bengal brands are spending significantly on marketing and brand building. Mukherjee said the new generation is doing it with real confidence — Manyavar, Emami, Bandhan, Bisk Farm and newer names like Joy and Dot & Key. “But what they should do more is simple: Stop treating marketing as only advertising. Budget for the whole journey, not just the message, for what happens after the customer buys, because that’s what keeps the promise. And build the brand, not just the distribution,” he added. “Trust is Bengal’s inheritance; this generation’s job is the ambition to take it national without spending away the authenticity that made it worth trusting,” he summed up.

The cockroach brief: Why virality is not a marketing strategy

The insect, as any entomologist will confirm, survives everything. Nuclear winters. Geological epochs. Even bad marketing strategy

DIGITAL
GADFLY



SHUBHO SENGUPTA

For roughly two decades — ever since “viral” migrated from epidemiology into digital marketing around 2007 — I have been the most unpopular person in the room whenever the word comes up. Clients demand it. Agency teams promise it. Award juries celebrate it with trophies that look increasingly like Toblerone chocolates. The best campaigns prove my point quietly. Ariel’s ‘Share the Load’ went viral and moved market share — because the virality served a precise commercial and cultural goal. Zomato trends regularly — because every piece of content points toward one measurable outcome: App opens. When virality serves a goal, it is a powerful amplifier. When virality is the goal, it is — as the Cockroach Janta Party demonstrated so elegantly — an expensive and exhausting way to go nowhere. Though, to be fair to the cockroach, it did get to CNN. In May, a senior judicial figure allegedly compared sections of unemployed youth to “cockroaches” and “parasites of society” in open

court. By the following morning, Abhijeet Dipke — a 30-year-old political communications strategist and Boston University PR graduate — had founded, named and distributed the Cockroach Janta Party across Instagram. Twenty million followers in 72 hours. Coverage on CNN, BBC and Al Jazeera. Street protests at Jantar Mantar, complete with cockroach costumes and mock manifesto slogans. Marketing academics will be studying this particular case for years — not entirely for the right reasons, but enthusiastically nonetheless. As a piece of communications craft, viewed purely as a marketing exercise, it was breathtaking. Jonah Berger’s STEPPS framework — the most rigorous model of content spread available — identifies six conditions for virality: Social currency, triggers, emotion, public, practical value and stories. The CJP hit five of these with a precision most agency creative teams spend careers chasing and award show entries try reverse-engineering. Let’s deep-dive: **Social currency:** Sharing a cockroach meme signalled political awareness, irreverence and humour simultaneously, at zero



ANTENNAS UP. The Abhijeet Dipke-led ‘infestation’ in New Delhi

personal risk. It made the sharer look sharp without requiring commitment to anything. **Triggers:** The original remarks recirculated through news cycles for days, each repetition refreshing the outrage and re-exposing new audiences to the CJP brand without a single rupee of media spend. The movement did not need to sustain its own attention — the news ecosystem did it for free. This is what the industry calls earned media. **Emotion:** Anger and humour activated simultaneously in the same container — high-arousal emotions. Anger alone exhausts an audience. Humour alone trivialises a cause. Together, they are

nearly unstoppable — as every successful political cartoonist from RK Laxman onwards understood. **Public:** Cockroach masks/costumes at Jantar Mantar gave the movement a visual identity that cameras could shoot and audiences could decode instantly — absurd enough to be funny, specific enough to be unmistakable. **Stories:** The underdog reclaiming an insult is one of the oldest narrative structures in human communication. The movement did not merely respond to being called cockroaches. It adopted the name, built a brand around it and turned the act of reclamation into the content itself. The one missing condition was

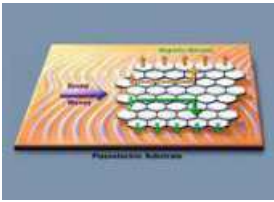
‘practical value’ — there was nothing useful to do with CJP membership beyond feeling agreeably righteous while scrolling. No conversion path. No next step. No answer to the question every successful campaign must eventually face: And then what? The only number that ultimately matters: 20 million Instagram followers produced a few hundred people on the street at Jantar Mantar. A conversion rate of approximately 0.001 per cent. No electoral registration, no policy platform, no formal political standing. The founder, it bears mentioning, was leading this revolution for unemployed Indian youth from Boston, Massachusetts, which is either deeply ironic or the most honest thing about the whole enterprise, depending on your appetite for contradiction. The cockroach, as any entomologist will confirm, survives everything. Nuclear winters. Geological epochs. Even bad marketing strategy. Whether this particular movement survives its own virality is, at this point, an open question. Virality without a goal is not strategy. It is noise with excellent metrics, a very good costume and absolutely nowhere to go.

The writer is a brand and communications strategist based in New Delhi

FROM THE LABS.

Using sound waves to bypass charge-based circuits

Researchers have discovered a new way to create and control magnetic information signals using sound waves. The technique



could help electronic devices use less power and generate less heat.

Today's electronics work by moving electric charges through circuits. This movement wastes energy as heat. As devices become smaller and faster, reducing this energy loss becomes increasingly important.

Instead of moving electric charge, the researchers use magnons — tiny waves of magnetic activity inside a material — to carry information. Magnons can travel with much lower energy loss than electrons.

The team from the Institute of Nano Science and Technology (INST), Mohali, showed theoretically that surface acoustic waves (sound waves that travel along the surface of a material) can create tiny distortions in a magnetic material. These distortions act like forces that push magnons and generate a spin current. This offers a new way to control magnetic information without relying on conventional electric currents.

The researchers built a theoretical model of an ultrathin magnetic material placed on a piezoelectric substrate. They found that sound waves travelling through the substrate can control the movement of magnons in the magnetic layer.

This work suggests that sound waves could become a practical tool for controlling magnetic signals in future low-power electronics, offering an alternative to conventional charge-based circuits.

Memory more suited for AI

Researchers in Germany have developed a new type of computer memory that could help make future electronic devices faster,



more energy-efficient and better suited for artificial intelligence applications. The technology, developed

jointly by researchers at Fraunhofer IPMS and semiconductor manufacturer GlobalFoundries, is based on ferroelectric random-access memory (FRAM). Unlike conventional memory devices, which may lose stored information when power is switched off, FRAM retains data permanently while consuming very little energy.

The new memory technology uses a material called ferroelectric hafnium oxide. Information is stored by shifting atoms within the material's crystal structure, creating tiny changes in electrical polarisation. Because this process requires very little power, the memory can operate below one volt and switch states within billionths of a second.

A key achievement of the project was integrating the new memory technology into GlobalFoundries' existing chip manufacturing process. This means the technology can potentially be produced on an industrial scale rather than remaining confined to the laboratory.

According to the researchers, the memory technology is particularly attractive for battery-powered devices, autonomous sensors and “edge AI” applications, where artificial intelligence runs directly on a device instead of relying on remote data centres. Lower power consumption could allow smart devices to process more information locally while extending battery life.

The researchers say the development combines an ultra low-power chip platform with a memory technology that is both fast and durable. They believe it could help support the next generation of electronics for applications ranging from industrial automation and automotive systems to medical devices and AI-enabled products.

● CLEAN TRANSITION

The time to deploy thorium is now

HOME ADVANTAGE. As demand outstrips uranium supply in the near future, India's vast thorium resources will come into their own

Anil Kakodkar

India's vast thorium resources offer a means for the country to become a global clean energy provider rather than remaining a major energy importer with the attendant vulnerabilities.

The founder of the atomic energy programme in India, Dr Homi Bhabha, recognised this way back in the early 1950s and had, in fact, chalked out a three-stage strategy.

The first stage comprising uranium reactors — essentially the pressurised heavy water reactors (PHWRs) — though delayed, has matured well. It is about to exceed the 10 GWe capacity envisaged for the beginning of the first stage.

Today, there is an unprecedented urgency for nuclear energy globally, driven by economic growth and the need to transition to clean energy. The new nuclear energy development in response, in India as well as globally, is essentially based on uranium. With the ‘once-through’ mode of uranium use that is in practice in most of the world, the known global uranium resources can support only around 500 GWe nuclear capacity, assuming a reactor life of around 60 years.

Against this, nearly three times larger nuclear capacity of 1,400 GWe by 2050, which is only 24 years away, has been projected by the World Nuclear Association. Clearly, resource depletion and supply-chain shocks are imminent within a decade or two. A shift to closed nuclear fuel cycles (reprocessing and recycling) and fast-breeder reactors, which can enhance the energy potential by around two orders of magnitude, is therefore inevitable.

The barrier to this shift are the proliferation concerns around plutonium. The way out of this conundrum is clearly thorium. It's



RIPE WITH OPPORTUNITY. India enjoys a lead in thorium as well as fast-breeder reactor technology A SHAIKMOHIDEEN

an opportunity India cannot afford to miss, given its lead in thorium as well as fast-breeder reactor technology, nor can it continue to suffer fuel supply vulnerability despite its vast thorium endowment and global shifts from fossil to nuclear energy.

The successful PHWR technology — the clear choice of investors — is expected to be the workhorse for the mission and may well contribute to at least half of the envisaged capacity. It offers an ideal platform to start irradiating thorium to produce uranium233 and directly connect it to the third-stage thorium reactors. This is necessary since the requisite thorium irradiation capacity through fast-breeder reactors is still decades away.

Thorium-HALEU fuel bundles, which are similar to the natural uranium fuel bundles used in PHWRs and can work with the existing design of PHWRs, can quickly enable thorium irradiation at scale along with significant economic,

safety and other dividends. The evolution of the second stage of fast-breeder reactors should continue in the meantime, exactly as per the original plan.

Fast-breeder reactors are needed to continuously augment India's nuclear energy resource at least until fusion reactors become a reality and deliver energy at the required scale. Accelerator-driven subcritical systems (ADSS) could also supplement fast-breeder reactors in meeting these objectives.

KEY CHALLENGES

The original three-stage strategy thus remains essentially the same, except for advancing the third stage to immediately follow the first. As a result, we should be able to leverage energy from the uranium233 derived from thorium in time before the challenges in the uranium supply chain start kicking in.

Dealing with the thorium fuel cycle has its challenges. The key is-

sue is with the hard gamma emitting daughter products of uranium232 that invariably accompany uranium233.

While that brings in proliferation resistance in the fuel cycle, it also greatly hinders the manufacture of uranium233-based solid fuel.

Fuel in liquid form, on the other hand, would seem more manageable, particularly since we can draw on our well developed capabilities in reprocessing and high-level radioactive waste immobilisation as well as benefit from insights on the global development of molten salt reactor (MSR) systems.

Unlike traditional nuclear engineering, which is physics-heavy, MSR engineering is chemistry-heavy. Much work is required in materials development, as well as the operational chemistry control of flowing liquid salt to prevent it from dissolving the reactor's pipes.

Intense co-ordinated work is needed to develop a thorium mol-

ten salt reactor (TMSR), which to my mind is a clear choice for the thorium stage.

Since thorium breeds even in the thermal neutron spectrum, TMSR operating in the near thermal spectrum could maximise sustained thorium-based power generation capacity, leveraging the spent thorium-HALEU fuel arising from a given PHWR capacity.

Instead of just jumping on the prevailing small modular reactor (SMR) bandwagon, SMRs based on TMSRs offer a safer and sustainable solution.

An intense R&D mission to advance the third stage and deploy TMSRs within 10-15 years is a critical national necessity.

In the near term, India must accelerate nuclear capacity build-up based on technologies that are already mature and by leveraging all deployment opportunities.

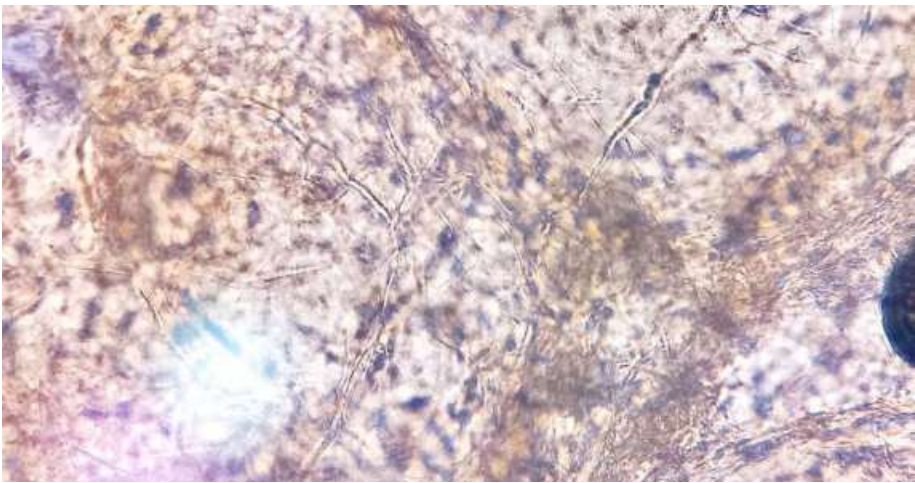
Captive clean electricity supply, leveraging brown field opportunity at retiring coal plant sites, clean hydrogen production — both through thermochemical splitting of water as well as electrolysis, clean base load energy for remote areas, and meeting diverse energy demands through micro-nuclear reactors are but some examples of the diverse opportunities that exist.

Getting the country ready to launch the third stage of thorium reactors in time should be done in parallel to making our energy security import-independent. A conducive ecosystem that can unleash Indian ingenuity to leverage the opportunities opened up by the SHANT Act has become the need of the hour.

The writer is a former Chairman, Atomic Energy Commission

We value your feedback. Do send your comments to quantum@thehindu.co.in

● WOW. Subterranean forest of fungi



Team Quantum

Imagine a forest so vast that if you could pull it out of the ground and stretch it into a single line, it would measure one-tenth the diameter of the Milky Way galaxy. Such a forest exists beneath our ground. Only, this forest is made not of trees but of fungi.

Scientists have just produced the first global map of an immense underground fungal network that quietly supports most of the world's land plants.

The fungi form microscopic thread-like structures that weave through the soil, linking themselves to plant roots in a remarkable partnership; the fungi supply vital nutrients such as nitrogen and phosphorus, while the plants pay them back with carbon-rich sugars produced through photosynthesis.

The scale of this hidden system is staggering.

Researchers found that nearly 40 per cent of the world's fungal biomass is packed into just the top 15 cm of soil in certain grasslands, especially high-altitude and seasonally flooded ecosystems such as Florida's Everglades. These underground fungal highways help lock away enormous amounts of carbon, making undisturbed grasslands some of the planet's most dependable carbon sinks.

The fungi are voracious carbon collectors. One estimate suggests they absorb around 4.3 billion tonnes of carbon dioxide equivalent every year — roughly 11 per cent of humanity's fossil fuel emissions in 2021.

But there is a worrying twist. The new map shows that many farming practices are tearing apart this hidden world. Topsoil in croplands contains, on average, only about half the fungal density found in healthier ecosystems — yet another instance of humankind destroying itself.

● LEARNING LEGALESE

AI aides to decode Indian law

N Nagaraj

NyayaAI, an AI-powered legal assistant developed by researchers at UPES Dehradun, is designed to answer legal questions in plain language. A lawyer researching a case precedent, a law student navigating a constitutional provision or a member of the public who cannot afford legal consultation can type a question on topics ranging from tenancy rights to criminal procedure and contract disputes, and receive a referenced answer.

The system can also summarise lengthy documents, identify relevant precedents and generate preliminary drafts of legal documents.

These developments have been captured in a paper titled ‘NyayaAI: An AI-powered legal assistant using multi-agent architecture and retrieval-augmented generation’, published in arXiv.

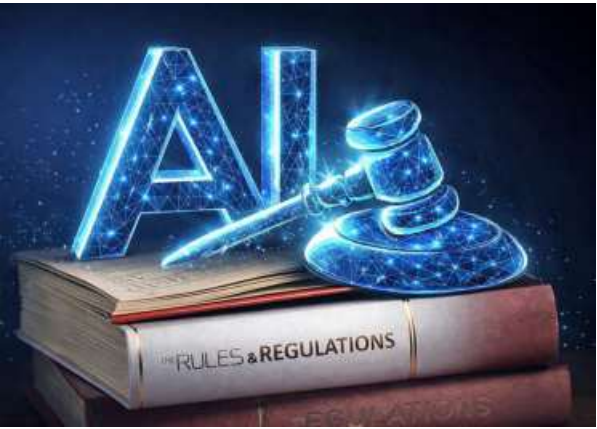
Legal help in India remains out of reach for most citizens. Consultation costs are prohibitive for many and legal services are concentrated in cities. The statutes are in English, a language in which the majority of Indians are not fluent. India has about two million registered advocates, most of whom are in urban centres.

Existing platforms such as Manupatra, SCC Online and Indian Kanoon operate on keyword search across digitised legal documents. While paid platforms have been adding AI-assisted search in recent years, none offers the integrated multi-agent task handling or RAG-grounded response generation that NyayaAI provides. AI legal assistants available internationally, including Harvey AI, CoCounsel and LexisNexis Protege, are built around American or British law.

BEYOND KEYWORDS

NyayaAI addresses a documented failure mode in legal AI: Systems relying on training data alone have generated responses citing cases that do not exist, leading to sanctions against lawyers in US courts.

Rather than generating



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from memory, the NyayaAI system searches a vector database of Indian legal documents using semantic similarity matching, retrieving provisions and precedents relevant to the query's meaning rather than its keywords, before generating a response grounded in those documents.

The system uses a multi-agent architecture orchestrated through the Mastra TypeScript framework. A main agent classifies incoming queries by legal domain, covering constitutional, criminal, civil, family and corporate matters, and routes complex queries to specialist sub-agents handling legal research, document summarisation, case analysis and drafting assistance. The drafting sub-agent generates preliminary versions of legal documents, including petitions, notices and agreements, which lawyers or users can review and modify. A compliance module intercepts all responses before delivery, validating them against legal, ethical and jurisdictional criteria and appending disclaimers where needed.

SUCCESS RATES

In evaluations, the system achieved 72 per cent response accuracy and 74 per cent retrieval precision. Domain classification precision reached 70 per cent across the five categories.

Criminal law and constitutional law recorded the highest precision, at 75 per cent and 73 per cent, respectively, which the authors attribute to cleaner terminology and stronger knowledge base coverage.

Corporate law recorded

the lowest precision at 65 per cent. The reason is jurisdictional overlap: A question about a company's acquisition, for instance, may fall simultaneously under the Companies Act, SEBI regulations, RBI guidelines and FEMA provisions, each administered by a different regulatory body with its own rules. The system must determine which framework applies before it can retrieve the right documents.

CAUSES OF FAILURE

Error analysis identified legal jargon complexity as the primary source of failure, accounting for 35 per cent of errors, given that the system is designed partly for users without specialist legal training. Jurisdictional ambiguity contributed 28 per cent, reflecting the difficulty of handling queries that span multiple regulatory regimes. Context misunderstanding, where the system responded to a different question than the one asked, contributed 22 per cent. Out-of-domain queries accounted for the remaining 15 per cent.

The paper's authors identify several directions for future development, including integration with the Supreme Court judgement repository and the India Code portal, multilingual support for Hindi and other regional languages, real-time update pipelines to keep the knowledge base current with evolving legislation, predictive analytics for legal outcomes based on historical case law, and support for user-uploaded documents to enable AI-assisted review of case-specific materials. The code has been made available to the public.

● FINANCING POWER

How the US funding cut impacts cancer research

Study captures the extent to which the fortunes of medical advancement is dependent on one region

K Bharat Kumar

The detrimental effect of the US government's move to cut funding for global healthcare programmes is widely known, but research now shows the quantum of the decline in funding.

A paper published by Anbang Du of the University of Southampton, UK, and others shows how medical advancement may be impacted if the US were to, hypothetically, end funding for all academic research. Moreover, the US is home to not only the largest R&D hubs globally but also major funders of research, and leads in scientific publication, intellectual property and clinical trial capacity.

The study shows that the government change in the US in 2024-25 was followed by an estimated 40.6 per cent reduction in central US funding for research agencies such as the National Institutes of Health (NIH) in 2026. It looked at 1,07,955 funding



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awards totalling \$51.4 billion globally.

The researchers modelled the cancer science ecosystem across a five-layered network, covering 233 countries and territories. The layers include grants, clinical trials, publications, inventions and patents. They found that the inequality between the West and low- and middle-income countries (LMICs) was most evident at the monetary end of the chain — namely grants and patents.

“The existing capacity in the research and innovation ecosystem

might limit LMICs' ability to acquire sufficient funding from international funders and commercialise the knowledge,” the report says.

Countries such as the US and China dominate all five layers and are, hence, at the core of the cancer science network, while smaller economies only partake of the paper-related layer and are seen as peripheral.

Low-income countries participate actively in the publication layer, but their presence is significantly diluted when it comes to clinical tri-

als, inventions and patents. The study has also come up with evidence of ‘tokenism’ in paper publication — when it comes to prestigious first- or last-name authorship positions, the involvement of researchers from low-income countries dips sharply, which also impacts their ability to generate research income.

Ultimately, this leads to a ‘brain drain’ — researchers tend to migrate to richer nations that can potentially offer them key roles in funding and capital creation.

The study postulates that the hypothetical removal of the US from the system causes a dramatic 50-plus per cent loss of efficiency.

What is the way out? The researchers suggest that other countries should scale up to make up for the decline in US funding. The largest contributions across the grants-to-inventions lifecycle will have to come from the European Union. They also suggest how other regions such as BRICS, which includes India, should chime in as well.

SEA CHANGE

Five new Acts to accelerate growth in the Indian maritime sector

1 MERCHANT SHIPPING ACT, 2025

- Robust framework for India's Maritime Vision 2047
- Ease of doing business and IMO-aligned regulations
- Welfare and skill development of seafarers
- Focus on safety, navigation and marine environment

2 COASTAL SHIPPING ACT, 2025

- Cost-efficient and green transport mode
- No licence for Indian vessels in coastal trade
- Strategic synergy with inland waterways
- Public database for transparency and investments

3 INDIAN PORTS ACT, 2025

- Integrated and evidence-based port development
- Centre-State consultative planning framework
- Mandates State maritime boards and dispute resolution
- Globally aligned safety, security and pollution prevention

4 BILL OF LADING ACT, 2025

- Simplified and modernised law replacing pre-Independence Act
- Clarity on rights, liabilities and transfer of ownership
- Reduces litigation and fosters trade efficiency
- Positions India on par with global maritime standards

5 CARRIAGE OF GOODS BY SEA ACT, 2025

- Replaces 1925 colonial-era law
- Retains Hague-Visby rules for global alignment
- Boosts India-UK CETA and global trade practices
- Enhances transparency and commercial efficiency

Illustration: Partha P Sharma Source: Ministry of Ports, Shipping and Waterways

GLOBAL BYTES.

Attack on seafarers condemned

BIMCO, the International Chamber of Shipping, INTERCARGO and INTERTANKO have unanimously condemned the recent attacks on seafarers and commercial shipping in and around the Strait of Hormuz. The International Maritime Organization says it has verified 46 attacks on international shipping in the region since February 28, with 14 confirmed seafarer fatalities. "There is grave concern about the recent escalation of the conflict, which puts the lives of our seafarers at risk, compromises the safety of international shipping and restricts freedom of navigation... Crews are civilian in nature and... often have no say regarding which cargoes are transported or the routes taken," says a joint statement.

AI-based alarm to keep wildlife off tracks

Alstom, a global leader in smart mobility, together with Flox Intelligence, is conducting field tests of AI technology to identify and deter wildlife along tracks. The tests show the system has potential to reduce the risk of collisions with wild animals — one of the common causes of disruptions on Swedish railways. AI-powered cameras identify animals in real time and activate tailored audio signals to scare them off the tracks. In the first test phase, the AI identified several species along the tracks, including moose, roe deer, fox and wild boar. The second implementation phase started in April to include both video detection and sound deterrence. Every year, around 5,000 animal collisions are reported in Sweden.

RE POSTS.

- ✕

India's maritime sector is becoming more inclusive than ever. The number of women seafarers has surged from 1,599 in 2021 to 7,033 in 2025, marking an impressive 340% growth. This milestone highlights the expanding opportunities for women at sea and India's commitment to building workforce.

Ministry of Ports, Shipping and Waterways
@shipmin_india
- ✕

India's aviation sector has witnessed remarkable growth over the years. From 74 operational airports in 2014 to 165 airports today, the nation has significantly expanded its aviation infrastructure, bringing more regions into the air connectivity network.

By connecting cities, enhancing accessibility, this transformation is driving economic growth, boosting tourism and strengthening India's position as a global aviation hub.

Airports Authority of India
@AAL_Official

Truck rental rates

Rentals for 18-tonne payload

Truck route	Rentals as on		% change
	May 28	Jun 11	
Delhi-Mumbai-Delhi	1,73,000	1,74,000	0.57
Delhi-Nagpur-Delhi	1,56,000	1,57,000	0.64
Delhi-Kolkata-Delhi	1,76,000	1,77,000	0.56
Delhi-Guwahati-Delhi	1,94,000	1,96,000	1.02
Delhi-Hyderabad-Delhi	2,00,000	2,01,000	0.50
Delhi-Chennai-Delhi	2,15,000	2,17,000	0.92
Delhi-Bengaluru-Delhi	2,21,000	2,23,000	0.90
Delhi-Ranchi-Delhi	1,65,000	1,67,000	1.20
Delhi-Raipur-Delhi	1,62,000	1,64,000	1.22
Delhi-Kandla-Delhi	1,40,000	1,42,000	1.41
Mumbai-Chennai-Mumbai	1,72,000	1,73,000	0.58
Ludhiana-Hyderabad-Ludhiana	2,00,000	2,02,000	0.99
Mumbai-Kolkata-Mumbai	1,97,000	2,00,000	1.50
Chennai-Ahmedabad-Chennai	2,11,000	2,12,000	0.47
Bengaluru-Kolkata-Bengaluru	2,15,000	2,13,000	-0.94
Bengaluru-Mumbai-Bengaluru	1,44,000	1,45,000	0.69
Delhi-Thiruvananthapuram-Delhi	2,83,000	2,82,000	-0.35
Guwahati-Mumbai-Guwahati	2,83,000	2,84,000	0.35
Nagpur-Chennai-Nagpur	1,63,000	1,64,000	0.61
Kolkata-Guwahati-Kolkata	1,55,000	1,56,000	0.64
Indore-Delhi-Indore	1,46,000	1,47,000	0.68
Ahmedabad-Indore-Ahmedabad	1,00,000	1,01,000	0.99
Vijayawada-Mumbai-Vijayawada	1,62,000	1,64,000	1.22
Vijayawada-Kolkata-Vijayawada	1,54,000	1,55,000	0.65
Mumbai-Nagpur-Mumbai	1,13,000	1,14,000	0.88
Mumbai-Pune-Mumbai	65,000	66,000	1.52
Mumbai-Aurangabad-Mumbai	83,000	84,000	1.19
Mumbai-Nashik-Mumbai	65,000	66,000	1.52

Figures in ₹ Source: Subham Freight Carriers India Pvt Ltd

CIRCULAR COMMERCE

Credit note buoys up ship recyclers

NEW FOR OLD. As the country's first ship recycling credit note goes out, domestic yards hope it will lead to more business

Aneesh Phadnis

India's ambitions for sustainable maritime growth received an impetus with the sector regulator issuing the first ship recycling credit note. On May 28, the Directorate General of Shipping issued the credit note to a Liberia-based shipping firm under a scheme to promote safe recycling and shipbuilding.

India is a global leader in ship recycling, handling about 30 per cent of dismantled vessels. The credit note scheme is expected to boost the competitiveness of Indian yards, attract fresh investments and strengthen the circular economy. Ship owners will get a credit note equivalent to 40 per cent of scrap value when they recycle the vessel in certified facilities. The note will be valid for three years and can be redeemed up to five per cent of the fair price of a new vessel built in an Indian shipyard.

"The ship recycling credit note scheme is a step in the right direction and demonstrates the government's intent to position India as a hub for ship recycling and shipbuilding," said Deepak Shetty, former Director General of Shipping. The scheme's launch also aligns with the implementation of the Hong Kong International Convention for the safe and environmentally sound recycling of ships.

The binding global treaty, which came into force last June, mandates recycling practices that minimise health risks and other occupational hazards for yard workers and prevent pollution through the proper management of hazardous materials and waste. Recycling yards are also required to facilitate worker training programmes, emergency preparedness procedures and environmental monitoring mechanisms.

"India currently has 111 ship recycling yards compliant with the Hong Kong International



THE BREAKDOWN. Ship owners will get a credit note equivalent to 40 per cent of scrap value when the vessel is recycled in certified facilities

Convention (HKC), at Alang, Gujarat. This represents one of the largest concentrations of HKC-compliant ship recycling facilities anywhere in the world," said Director General of Shipping Shyam Jagannathan. He said the Directorate, in coordination with the State authorities and recognised organisations, undertakes periodic inspections, audits and certification activities to ensure continued compliance.

A POSITIVE START

According to the Ship Recycling Industries Association (India), 119 ships were recycled in the country in FY2026, amounting to 1.08 million light displacement tonnes — an industry metric for measuring the quantity of recyclable material available for processing. This was marginally higher than the 113 ships recycled at Alang in FY25.

The recycling business has seen a slowdown with geopolitical conflicts pushing up freight rates. As such, fewer ships are coming to Alang for scrapping. The new scheme and the initiatives in digitalisation and ease of doing business will be an added support.

Jagannathan said the ship recycling credit note scheme has



Shyam Jagannathan, Director General of Shipping

seen encouraging traction. Already 41 ship owners have registered, and six applications have been received under the scheme. The first credit note, valued at ₹29 crore, was issued for recycling a St Kitts and Nevis flagged capesized vessel, called *Kosta*. The ship, owned by Bella Shipping Inc, was recycled at the Inducto Steel Limited yard at Alang in April.

"All ships recycled at HKC-compliant yards in India are eligible for the credit note, subject to adherence to the scheme's prescribed conditions and procedures," Jagannathan said.

A unified ship recycling portal enables seamless applications, processing, tracking and issuance of digitally signed credit

DISTRIBUTION MIGHT

How logistics will package and deliver demand growth



BALFOUR MANUEL

India's demand story is typically narrated through income. Wages, remittances, rural credit: These are the variables that dominate the macro conversation, and there's nothing wrong in this. But the more revealing divide emerging across the country's demand map is not 'rural versus urban'. It is 'reachable versus hard-to-serve'. Private final consumption expenditure grew 7 per cent in Q1 FY27, led by resilient rural demand. The constraint on that growth is no longer production capacity but distribution certainty. Distance, more than income, is quietly determining where consumption can actually occur, and the story of the next demand cycle begins there.

This shift sits within a longer, deliberate policy arc. Tax harmonisation has reduced internal trade friction, economic corridors have improved line-haul predictability and digital systems have strengthened shipment visibility. The country's ranking on the World Bank's logistics performance index improved from 44th in 2018 to 38th in 2023, out of 139 countries. The two dedicated freight corridors, now operational across 2,843 km, run through Uttar Pradesh, Bihar, Rajasthan and Gujarat — States where distance had long constrained broader market access.

Global supply chain disruptions have raised the stakes for the domestic logistics argument. Companies previously treated geopolitical risk as a temporary disruption, carrying buffer stock until familiar routes reopened. That assumption has not held. Lead-time predictability has shifted from a procurement footnote to a primary selection criterion. What began as cost arbitrage has quietly become resilience arbitrage. Cost arbitrage is comparatively simpler to model. Resilience arbitrage demands net-



IN EVERY NOOK. Reliable supply births new consumption centres ISTOCK

work depth, freight route diversity and end-to-end logistics dependability, even under pressure. The gap between India's manufacturing reputation and logistics reliability directly determines whether the supply chains that have already shifted stay shifted.

Inside India, those pressures are redrawing the domestic demand map. Distance once functioned like an invisible tariff. Companies carried excess safety stock because replenishment was uncertain, and expansion into smaller markets was sequenced cautiously. That posture is easing.

PREDICTABILITY MATTERS

For small manufacturers and farm-gate producers, for whom thin margins leave no room for delayed receivables, predictable delivery changes the unit economics of expansion. Dedicated freight capacity is reducing congestion on critical routes, while expressways support predictable overnight movement between manufacturing clusters.

Firms are entering emerging consumption centres not because those markets have suddenly become lucrative, but because they have become reliably serviceable. India is enlarging the serviceable zone, not just its consumer base.

That enlargement makes logistics a revenue condition, and not

merely cost discipline. When replenishment cycles stabilise, production planning firms up. Capital released from buffers moves into deeper distribution.

Yet infrastructure efficiency on paper does not automatically translate into service on the ground. Urban congestion, land constraints and driver availability remain daily operational realities.

The corridor-to-last mile gap remains wide enough to limit gains in underserved markets. Logistics alone cannot substitute for income growth. But it can accelerate the expression of income. When friction between purchase intent and product availability declines, latent demand surfaces earlier than macro data predict.

The next consumption cycle may not announce itself through a sharp rise in income. It is more likely to arrive as a quieter shift, in the markets that have become serviceable, in the categories where purchase intent no longer waits on proximity.

The decisive question, it turns out, was never just about income. It was also always about reach.

The writer is Managing Director, Blue Dart

We value your feedback. Do send your comments to bl.logistics@thehindu.co.in

TECH TRAIN

RITES taps AI to step up service and revenue



ON TRACK. RITES uses AI to fine-tune rail inspection

Chitra Narayanan

Every organisation is scrambling to use AI and other tech to improve productivity. At RITES — Rail India Technical and Economic Service — the attempt is to improve service quality and revenue. "We have recently launched a new centre of digital excellence and innovation hub, called Abhikalp, to speed up the use of tech and AI in our services," says Rahul Mithal, CMD of RITES

He says the integration and adoption of AI and other tech at the PSU goes beyond mere housekeeping. "Each of our 13 verticals has been given a target to find out tech and AI enablers, quantify how it is resulting in better service, and measure the improvement in terms of the percentage of revenue."

The infrastructure consultancy company's 13 verticals include railways, ports, airports, metro system, highways, ropeways, and bridges; it offers three broad areas of service — techno-feasibility studies, project management consultancy and third-party audits. It is currently executing 700 orders across these verticals

On the new tech push, Mithal says, "Let's say you are one of the 13 verticals, and your annual revenue is hypothetically ₹100 crore, then we are asking what percentage of revenues this year, in 2026-27, will be the add-on because of a tech-enabled intervention."

Mithal says an AI-based application that has become hugely popular within the organisation is in the domain of rail inspection.

"We have been doing this historically for five decades — safety inspections of rails, which is very critical for Indian Railways. We developed a tool called VISTAR — Visual Inspection System through AI for Rails — and deployed it last year to assist inspectors," he says. The tool leverages AI and machine learning to drive precision in the quality checks.

As Mithal explains, VISTAR collates the historical data related to vulnerable areas, and then provides the inspectors with the precise data to zoom in directly to the problem spots.

"This is exactly like the logic doctors use — they look at past clinical patient history records and then suggest interventions, trying to pre-empt ailments. We are also giving a template, so the inspectors can do a more efficient check of the rails," says Mithal.

Not only does this cut time, but the quality of inspections has also radically improved with a focus on the prevention of adverse incidents, he says.

MY FIVE.

DEVYANI JAIPURIA
Director, RJ Corp – Healthcare
& Education



Tennis, sustainable meals and painting as a quiet counterpoint

- 1 Movement:** I start early, before the day gathers pace. Movement helps me settle into the morning — strength training, stretching or, when possible, a short game of tennis.
- 2 Nutrition:** My meals are simple and home-cooked, whenever possible. Fresh ingredients and seasonal foods work best, especially on long and unpredictable workdays. I eat in a way that feels sustainable rather than restrictive.
- 3 Balance:** Routine builds balance. When the body is active and the mind has clarity, decision-making becomes steadier and the day feels more manageable.
- 4 De-stress:** Tennis helps release physical tension and clears my head. Painting offers a quieter counterpoint. Even carrying a sketchbook allows me to pause briefly and notice details I might otherwise miss.
- 5 Winding down:** Evenings are about easing out of the day's momentum. After dinner, I unwind with my husband over a film or a few pages of reading. Creating a clear break from work helps me return the next day with better focus.

CAPSULES.

Undercutting AIDS response

A recent UNAIDS report shows that external funding cuts, a strong pushback on human rights, and under-investment and under-prioritisation of HIV prevention and community services are threatening to reverse years of gains in the AIDS response. “There’s no question that this is the most serious disruption in the HIV response since the world came together to fight this disease,” said Winnie Byanyima, Executive Director of UNAIDS. “The funding cuts, combined with the reduction in civic space and the further criminalisation of marginalised populations have come together to create the biggest storm the HIV response has ever seen.” Dramatic cuts in aid that highly burdened, low-income countries depend on for their HIV response have had a devastating impact. Global development assistance from multiple countries fell by 23 per cent in 2025 — the sharpest drop on record — and HIV programmes have been hit hard, the agency said. HIV prevention is being dismantled at the very moment the world needs to take it to scale, especially with new, revolutionary, long-acting prevention innovations coming to market, the note said. HIV response has been the most successful story in global health over the last 25 years, as AIDS-related deaths have been reduced by 56 per cent — from 1.3 million in 2010 to 5,70,000 in 2025. But this success is fragile — nearly 9 million people are not on treatment.

COMING UP.

Ageing healthy with yoga

This year marks the 12th International Day of Yoga (June 21), and the theme ‘Yoga for Healthy Ageing’ emphasises the importance of yoga for all ages, a note from the United Nations said. Yoga can be a valuable practice for healthy ageing because it supports many of the abilities that help people stay active, independent and engaged as they grow older. By combining gentle movement, stretching, breathing exercises and mindfulness, yoga can help improve balance, flexibility, strength and mobility, while also supporting mental well-being.

OBESITY RISING

A widening crisis for women

NFHS-6. With 43% urban women now obese, data reveals a crisis with a distinctly female face

J Mangaiyarkarasi

Women in Indian cities seem to be carrying more than their share of the weight — quite literally. The latest National Family Health Survey (NFHS-6) data shows that nearly 43 per cent of urban women are overweight or obese — a steep climb from 33 per cent in the last survey. In short, the country is on the brink of a worrying demographic where half of all urban women aged 15–49 will have a body mass index (BMI) above 25. Men are adding the pounds, too, (rising from 28.6 per cent to 36 per cent), but the divide is widening, with women bearing the brunt of this expanding crisis. Obesity was once seen as a disease of affluence, but the numbers now tell a more complicated story, driven by changing food environments, rising chronic stress levels, metabolic diseases and the lingering effects of malnutrition. And as waistlines expand, so does the market for GLP-1 weight-loss drugs, which are flooding India after the expiry of semaglutide’s patent in March 2026, promising quick fixes to those desperate for relief. But experts caution that medication cannot outrun the structural drivers of obesity nor can it stand in for the deeper shifts needed in food systems and social norms. “Obesity is often associated with overeating, but it can emerge from poor, non-diversified diets too,” said Dr Rama Narayanan, Senior Fellow, Nutrition and Health, at MS Swaminathan Research Foundation. The trend, she said, is being driven by lifestyle changes linked to urbanisation and a shift from traditional millet-based diets to refined cereals and ultra-processed foods. In the case of women, Dr Narayanan said, the vulnerability factors are compounded by what time-use surveys call the ‘poverty of time’. “Women spend long hours on care work in addition to



A SONG AND DANCE. Doctors in Ongole shake a leg to spread awareness on healthy food habits to curb obesity KOMMURI SRINIVAS

Overweight or obese adult Indians (BMI ≥25)

	Women			Men		
	Urban (%)	Rural (%)	Total (%)	Urban (%)	Rural (%)	Total (%)
NFHS-4 (2015-16)	31.3	15.0	20.6	26.6	14.3	18.9
NFHS-5 (2019-21)	33.2	19.7	24.0	28.6	18.8	22.9
NFHS-6 (2023-24)	42.8	25.5	30.7	36.3	23.0	27.3

Source: NFHS

paid employment. Chronic stress, lower physical activity, poor sleep and reduced access to nutritious food all combine to increase obesity risk,” she explained. Rural India is not far behind either, as ultra-processed foods seep into villages (see *graphic*). And that becomes a cause for concern, say experts, as obesity is a major risk factor for diabetes, hypertension, cardiovascular disease and several cancers. **LINGERING COVID EFFECTS** The first National Family Health Survey after the Covid-19 pandemic, the NFHS-6 captures the aftermath of lifestyle changes from the lockdown era — reduced mobility, unhealthy routines and higher stress leading to more weight gain, said Dr Yashendra Sethi, cardiovascular researcher and editor, *BMC Public Health*. The rise of health fads driven by social media influencers (often not medically qualified) has added another layer of worry. Sethi noted that “women, with a higher baseline body fat percentage, are particularly affected because the structured leisure-time exercises remain logistically and socially harder”. Vishnu N Ramani, Health Economist at PHFI Institute of Public Health Sciences, agrees. In a country where many work in the informal sector, most women don’t have the privilege of going to a gym or for a walk or exercise in an outdoor park, he said, point-

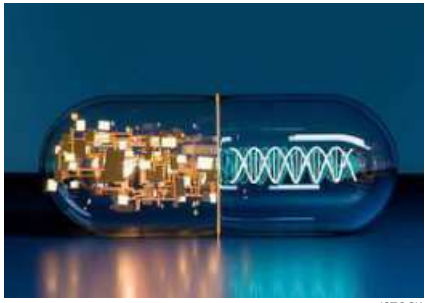
ing to the cultural and social dynamics of gender shaping the obesity crisis. “A woman won’t get much time to go for exercise because she has to look after her domestic home,” he explained. Then there are the commercial determinants of health. Ramani observed that high-calorie foods have become cheaper over the years, and more accessible compared to healthy foods. The rise of the platform economy has also reshaped food choices across income groups, accelerating the penetration of ultra-processed foods in urban and rural India. India’s obesity story differs from that of many fast-developing countries, as it faces “an overweight and obesity pandemic before undernutrition has settled”, added Sethi.

‘TRIPLE BURDEN’

Public health expert Dr Nirupama AY, who is with Hyderabad’s Indian Institute of Public Health, pointed to the Developmental Origins of Health and Disease (DOHaD) theory to explain that babies exposed to undernutrition during the mother’s pregnancy or in their infancy can develop metabolic adaptations that favour energy storage. When they later encounter improved diets and greater calorie availability, they may be more prone to storing fat and developing obesity, even without excessive food intake, she explained. “The population that was undernourished 20 years ago is now becoming obese... that’s where the numbers come from,” Nirupama noted. Experts said India’s “triple burden” of malnutrition, obesity and undernutrition reflects a complex development trajectory, where old nutrition challenges have not disappeared even as newer ones emerge. Dr Narayanan called for the inclusion of more pulses, millets and healthier oil alternatives like groundnut oil in public distribution systems — providing the micronutrients and proteins needed in one’s diet, rather than carbohydrate-heavy options that feed the very crisis policymakers are trying to fight.

FROM THE REGULATOR

UK sandbox for AI health innovations



ISTOCK

Patients across London are set to benefit from faster access to innovative, AI-powered healthcare technologies, as the Medicines and Healthcare products Regulatory Agency (MHRA), in partnership with NHS England (London) and London Health Innovation Networks, announced a new regulatory sandbox. The initiative, known as London Region I, will create a real-world environment to safely deploy AI-enabled medical devices while generating additional evidence of benefit across NHS settings, helping ensure patients can access new technologies earlier while maintaining the highest standards of safety. By bringing together regulators, healthcare providers and innovators, the sandbox provides a controlled environment to safely deploy cutting-edge technologies. The programme will support the delivery of the NHS 10-Year Health Plan by enabling faster adoption of technologies that improve patient outcomes, expand access to care and reduce health inequalities. Up to 10 AI medical device manufacturers will be selected to participate in the initial phase, working alongside NHS providers across London to deploy technologies in live clinical settings under MHRA oversight. By enabling controlled, outcomes-based deployment, the sandbox aims to generate robust real-world evidence on safety and effectiveness while supporting a clearer, more predictable route to wider adoption. MHRA will be inviting expressions of interest from both NHS providers and AI medical device manufacturers interested in participating in London Region I next month. The programme will support collaboration between providers and innovators, helping to match technologies with real system needs and enabling rapid, evidenced deployment in practice. Lawrence Talon, MHRA Chief Executive, said, “This initiative demonstrates that regulation can be an enabler for innovation, not a barrier. We need to work faster to keep pace with developments in AI to ensure that patients can safely benefit from cutting-edge technologies as they become available.”

Source: MHRA

LENS ON HEALTH



STRENGTHENING HANDS. Congolese healthcare workers participate in an accelerated training session at newly built tents by the Samaritan’s Purse International Relief at the Scott Powell Memorial Ebola Treatment Centre as aid agencies intensify efforts to contain the outbreak caused by the Bundibugyo virus in Bunia town, Ituri province, Democratic Republic of Congo REUTERS

Medical negligence: Fair redressal can restore trust

BY INVITATION



ALEXANDER THOMAS

A public interest litigation (PIL) filed before the Supreme Court for the removal of doctors from the ambit of the Consumer Protection Act has generated considerable debate. Unfortunately, some have misunderstood its intent. The objective is not to reduce accountability for doctors, but create a grievance redressal system that serves patients better, resolves disputes more quickly and restores trust between patients and healthcare providers. Trust is the foundation of the doctor-patient relationship. Patients entrust doctors with their health, well-being and, often, their lives. When outcomes are unfavourable, most patients seek something basic — an explanation. They seek honesty, empathy and reassurance that their concerns are being heard. The current consumer court system has largely failed to meet these expectations. Thousands of cases are pending across the country, often for years, because of backlogs, vacancies and the complexity of medical evidence. This causes distress to patients and doctors alike, while eroding trust, underscoring the need for a more patient-centred system. For years, I have advocated for patient safety, including formal training in communication skills and medical ethics. Through educational initiatives, including the book *Communicate. Care. Cure.*, and engagement with regulators and professional bodies, these subjects have been incorporated into healthcare curricula in India. This is important because many disputes arise not from poor treatment but inadequate communication, unrealistic expectations and lack of empathy. I have also worked with organisations such as the Patients for Patient Safety Foundation, which prioritises giving patients and families a meaningful voice in healthcare and grievance redressal.

MEDIATION FIRST

The need is for strengthened and standardised hospital-based griev-



ISTOCK

ance redressal systems with patient representation. Many hospitals already have patient grievance committees endorsed by the National Accreditation Board for Hospitals & Healthcare Providers. Mediation should be central to this approach. When patients receive timely explanations, sincere communication and a fair review of their concerns, many disputes can be resolved without prolonged confrontation. Importantly, such a system would not take away any patient’s legal rights. If patients are dissatisfied with hospital-level grievance processes or mediation, they should have access to existing legal and regulatory forums. The objective is to provide an effective first level of redressal, not eliminate accountability. **DEMOTIVATION** The existing system also has unintended consequences. Fear of litigation encourages defensive medicine, increasing healthcare costs. The resulting atmosphere of mistrust has also contributed to violence against healthcare workers and hospitals. India must consider the long-term impact on the profession. Medicine has traditionally attracted some of the brightest minds in the country. If it becomes increasingly associated with litigation, hostility and personal risk, fewer talented young people may choose it as a career. The goal should be improved accountability that is transparent, fair, timely and patient-centred.

The writer is Founder and Patron, Association of Healthcare Providers, India. Views are of the association

‘The kindergarten fitness class for the 50-plus’

An online yoga initiative that focuses on staying the course, across age groups and fitness levels

PT Jyothi Datta

“I just want more people to become regular with practice,” says Saurabh Bothra, the young IIT-ian who has brought yoga into many homes across the country. Bothra’s “habit building” online yoga initiative, Habuild, was baptised by fire, in a sense. It was founded in March 2020, just as the world was grappling with the Covid-19 pandemic. His early sessions had three people. While that changed gradually, so did his approach to life: From wanting to teach yoga to more people to wanting people to become more regular with their practice, he says. “This small change takes a long time,” he adds. And it’s not about the yoga practice, he says. “My understanding of yoga is, if you do anything with discipline, that’s part of yoga practice. So, we are

teaching yoga, but not the conventional mindset people have that yoga is this asana, that asana, handstand, full split and all. That’s not how we approach it,” says Bothra, who will turn 34 this month. “We make it so simple that anyone and everyone can become regular with it,” he explains. A testimony to this inclusive approach is the nearly 80 lakh people who have done yoga with Habuild, with the oldest at 104 years, and the youngest at three, he says. The initiative is focused on those who have spent their lives looking after others, like “moms”, he says. But that also means an age group that may have multiple ailments. For those with critical ailments, he says, Habuild clarifies upfront that the team is not made of doctors, and what’s on offer is not treatment. They want to reach out to those who have never attempted any exercise before and are



DRINK TO HEALTH. Saurabh Bothra, founder of online yoga platform Habuild, and his mother Vijaya

away from fitness apps and so on. “Our job is like a kindergarten teacher, taking students who do not know ‘A-B-C-D’ to help them understand what H-E-A-L-T-H means... They can graduate and go to an offline class, a gym or another setup where there is more intensity. But we want to remain a primary school only, where people come, understand and, if they like it, they can stay back... the kindergarten class of health for 50-year-olds.” In fact, Bothra’s mother, Vijaya, the Chief Financial Officer, is also “chief critical officer”, he jokes, adding that her insightful feedback is always acted on. Habuild is looking to bring in multilingual formats, besides initiatives focused on nutrition and sleep, he says. Up

ahead, the initiative’s three pillars will be exercise, diet or nutrition, and sleep or rest. His team is 350-strong, including his fellow IIT-ian Co-founders Trishala (who is also his sister) and Anshul Agrawal. The venture has partnerships with corporates and the Central Ayush Ministry. “It took us one year to realise that we want to focus on moms and not engineering students,” Bothra says on the initiative’s growth trajectory. Soon they shifted gears, and the audience of three increased to “100 moms”, he says. As the initiative grew to 1,200 moms, slowly men joined too. The number spiked to 80,000 in 2023, post Covid, via WhatsApp referrals. Addressing the criticism that what’s on offer is not classical yoga, Bothra says he agrees. “I want to do what they want me to do,” he says, reiterating he was not trying to teach yoga but helping people become regular.

QUICKLY.

**Indian-flagged dhow
sinks off Oman coast**

Dubai/New Delhi: An Indian-flagged dhow sank off the Oman coast following mechanical failure and all 14 crew members were rescued on Sunday. The Indian Embassy said that all 14 crew members rescued are “safe and in good health”, adding that they had been moved to a vessel that was heading to Mumbai. The Mumbai-bound vessel, *MV Jabal Ali*, is learnt to be a cargo ship. “Rescue operation involving Indian flagged *MSV Virat 1* has been successfully completed. All 14 crew members have been rescued and are presently onboard *Jabal Ali 9* heading to Mumbai. The crew members are safe and in good health,” said the mission in a post on X. The incident comes amid American military action on merchant vessels defying the US’ naval blockade in the region. **PTI**

Restrict seafarer deployment in the Persian Gulf: Govt to shipping firms

SECURITY RISKS. Move follows attacks on vessels that tried to bypass the US-imposed blockade

Our Bureau
Mumbai

The Directorate General of Shipping (DGS) has asked fleet owners and recruitment firms to restrict deployment of seafarers in the Persian Gulf region. The move follows attacks on commercial vessels that have tried to bypass the US-imposed blockade in the Gulf waters. **SEAFARERS KILLED** Three Indian seafarers were killed in a US strike on oil tanker *MT Settebello* on Wednesday. Two other vessels manned by Indian crew — *MT Marivex* and *MT Jalveer* — also came under attack



LETHAL ATTACKS. Three Indian seafarers were killed in a US strike on oil tanker *MT Settebello* on Wednesday

last week. On Saturday, the DGS said all recruitment and placement firms and shipping companies had been advised to restrict the deployment of Indian seafarers to conflict areas. “However, companies may carry out crew change in emergency situations with

the consent of crew,” said the DGS in its security advisory. **‘STEP UP VIGILANCE’** In view of the heightened security risks, the regulator asked all ship masters (captains) to increase vigilance. “Masters of vessels operating in or transiting through the Gulf region, including the Strait of Hormuz and adjoining waters, are advised to maintain heightened security awareness, closely monitor navigational warnings received and advisories issued from security agencies, and implement all applicable ship security measures and company security procedures,” it said. According to the International Maritime Organization (IMO), there have been 46 attacks in and around the Strait of Hormuz since February 28, resulting in 14 casualties. IMO Secretary General Arsenio Dominguez has condemned the attacks on the vessels, calling them “simply unacceptable”. The External Affairs Ministry summoned US diplomats two times to lodge protest against the attacks and deaths of Indian seafarers. On Saturday, External Affairs Minister S Jaishankar spoke with US Secretary of State Marco Rubio to reiterate the government’s protest. “Such lethal actions against commercial shipping are not justified,” Jaishankar told Rubio.

FSSAI cracks down on misleading brand names and their claims

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New Delhi



In a crackdown on violation of labelling norms, the Food Safety and Standards Authority of India (FSSAI) has issued notices to several food companies for misleading brand names, trade names and product claims. The food regulator has issued notices to these players, stating that they are in violation of provisions of the Food Safety and Standards Act 2006.

BRANDS INVOLVED The brands include Healthy Master, Neuherbs True Vitamin, Plan B, The Health Factory, Troovy, Healthy Choice, Emami’s Healthy & Tasty and Health Aid. Notices were also sent to Organic Wisdom, Shine Organic, Two Brothers Organic Farms, Storia, World of Organic and Iota Water.

In a social media post on X, the FSSAI said: “These FBOs are directed to strictly comply with the established labelling and display regulations to prevent consumer deception.”

In its notices, FSSAI pointed out that the trade names of brands, including snacks brand Healthy Master, The Health Factory (Zero Maida Whole Wheat Bread and Zero Maida Pizza Base), Healthy Choice (Poha), Emami’s edible oil brand Healthy & Tasty and Health Aid (Salmon oil), are “likely to mislead the consumers”.

INGREDIENTS VARY It also stated that these trade names “appear to be in contravention of applicable FSSAI regulations”. In the case of Troovy’s Healthy Mix chips, the claim “healthy” was found to be misleading as it contains

The food regulator has issued notices to the players, stating that they are in violation of provisions of the Food Safety and Standards Act 2006

other ingredients. Meanwhile, in its notice to Neuherbs True Vitamin, it pointed out that the term “true vitamin” is neither defined nor recognised in the regulations and may mislead consumers. It noted that the trade name “Plan B plant-based vegan” creates the impression that the products are vegan, but the products have not obtained prior approval for vegan food and vegan food endorsements in their FSSAI licence.

ORGANIC STATUS In its notices to companies Organic Wisdom, Shine Organic and Two Brothers Organic Farms, the food safety regulator pointed out that these companies’ brand names were likely to mislead consumers regarding the organic status of the products. The FSSAI noted that these brands did not have the requisite certification, FSSAI Jaivik Bharat logo and the necessary organic endorsement.

Flipkart moves Kolkata HC for GST exemption on delivery charges

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E-commerce major Flipkart India has moved the Kolkata High Court with a plea to grant the company GST exemption on delivery charges by treating its services as those provided by a goods transport agency (GTA). This petition has been filed after the West Bengal Appellate Authority for Advance Ruling (WBAAAR) rejected Flipkart’s attempt to claim GST exemption by classifying itself as a GTA, setting aside a May ruling by the authority allowing the exemption. A writ petition by the e-commerce major, as seen by *businessline*, prayed for a declaration that “the services

supplied by the petitioner described herein qualify as goods transport agency services”. It also prayed for a declaration that GST exemption be granted to Flipkart under Entry 21A of Notification No 12/2017-Central Tax (rate) dated June 28, 2017. **WHAT IS GTA?** A GTA is any business that transports goods by road and issues a consignment note, the document showing that the transporter has taken responsibility for the goods until delivery. This particular provision exempts GTA services from GST when provided to an unregistered person, and is meant to offer relief to unregistered individual consumers from paying GST on road freight and transport

delivery. Flipkart’s step to move the High Court comes at a time when transport operators and SME groups have taken steps to appeal to various State-level commercial tax authorities to deny the exemption, taking a cue from the WBAAAR ruling. Flipkart did not respond to *businessline* queries on the GST exemption. **TN TAX REVENUE** K Narasimhan, Convener of the Forum For Progressive Gig Workers in India, told *businessline* that it has made representations to Tamil Nadu’s commercial tax authorities to prevent e-commerce companies like Flipkart and others from seeking GTA exemption and investigate the issue. Tamil Nadu’s tax revenue has been slow this year, and addressing this

issue could bring back close to ₹300-350 crore of GST revenue for the State. Authorities in Rajasthan and Punjab are also said to be seized of this issue. According to tax experts *businessline* spoke to, WBAAAR examined Flipkart’s business model and ruled that its integrated logistics operations do not qualify as GTA services. The WBAAAR noted that issue of a consignment note is not sufficient, and Flipkart is not just a courier but runs end-to-end process, where goods go through source hubs, sorting centres, transshipment points and last-mile delivery agents. Such operations, thus, attract 18 per cent GST on delivery charges, it ruled. According to those in the know, the new invoicing

structure, which considers the seller marketplace fees as goods transport agency charges, will lead to revenue and margin cuts for sellers, and the impact could be around ₹2,600 crore across small businesses and FPOs. Industry sources noted that this issue needs to reach a logical conclusion as early as possible. **TRADE DEALS** According to an executive with the logistics industry, India is building a globally competitive manufacturing and export economy, and is negotiating trade agreements that will open new markets. All that ambition travels in the end on the same road — the integrated digital logistics network that moves goods from producer to buyer, across every geo-

graphy and every border, said the executive. “That road must stay free,” the executive emphasised. “The recent ruling raises an important question for policymakers and tax authorities: whether the GST treatment of modern e-commerce logistics services is fully aligned with the nature of the services being delivered. As India’s digital economy continues to expand, regulatory clarity and uniform enforcement become critical. Any inconsistency in tax treatment not only has implications for government revenues, but also for the millions of MSMEs that compete within the same ecosystem and are expected to comply with the law in both letter and spirit,” said Vinod Kumar, President, India SME Forum.

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